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Model, Label, or Analyst? A Variance Decomposition of Real-Time Sepsis Onset Prediction on MIMIC-IV

By
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Abstract

Over ninety AI models have been built to predict sepsis in real time, yet they fail in practice. Wang et al. (2025) showed that while AUROC barely changes between internal and external testing (0.811 to 0.783), the clinical Utility Score collapses from +0.381 to -0.164 , meaning these models cause more harm than benefit when deployed. This thesis investigates why: is the problem the model, the sepsis label, or the researcher’s methodological choices?

The study uses MIMIC-IV v3.1 (65,078 adult ICU stays) with external validation on eICU-CRD v2.0 (63 sepsis cases across 7.9 million person-hours). Three Sepsis-3 label variants (Narrow, Seymour-Standard, Liberal) are tested with a dynamic discrete-time competing-risks hazard model, gradient-boosted trees (GBT), a static Cox model, and rule-based scores (NEWS2, qSOFA, SIRS). Because all three variants share one culture-plus-antibiotic anchor, the Sepsis-3 conjunction is additionally split into its two limbs and each is labelled separately, including a deterioration label that uses no treatment timestamp at all, and Sepsis-3 is relabelled a third time under the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ onset timing of Seymour et al. (2016) and the PhysioNet/CinC 2019 Challenge, which allows an onset to precede treatment.

The results invert the study’s own premise. Model class dominates the decomposition, moving AUROC by 0.1499 against the label definition’s 0.0123 — a difference of -0.1375 whose interval $[-0.1623, -0.1085]$ excludes zero. Yet the three labels disagree sharply about *which* patients are septic ($\kappa = 0.267$ between the two narrowest readings, at near-identical prevalence), so label choice sets the cohort and the denominator of every downstream clinical claim without much affecting how learnable that cohort is. 6 of 22 shared covariates are unstable across labels, including a sign reversal on vasopressor exposure. Random data splits do *not* inflate AUROC relative to time-based splits (-0.0165 , 95% CI -0.0372 – 0.0040), and discrimination transports to eICU-CRD without measurable loss (-0.0033 , 95% CI -0.0659 – 0.0676). An interpretable hazard model matches gradient-boosted trees to within -0.0129 AUROC $[-0.0272, 0.0019]$, inside the pre-registered equivalence margin. The pre-treatment evaluation window — the hours before the first antibiotic or culture — contains **zero sepsis events** in 179,003 person-hours. That

zero is proved here to be a logical consequence of the conjunction-based onset rule rather than a discovery about the data: when the onset of the *same* labelled stays is timed by the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ rule instead, 1,738 onsets (27.0%) move earlier and 71 **fall before treatment begins**, identified at 0.8081 AUROC. Sepsis-3 onsets can precede clinical action; it is the conjunction-based onset rule, widely used in this literature, that prevents them from being recorded. Sweeping the alert threshold rather than scoring at an arbitrary cut-off, **no model exceeds a Utility Score of 0.039** under the primary label — a few per cent above the strategy of issuing no alerts at all, on a scale whose upper bound is 1 — and that maximum is bought at 44.4 false alerts for every true one, against a benchmark of 1.4. Under that label four of the six models have no threshold at which alerting beats silence at all.

Splitting the definition shows how much of the label effect is definitional and how much is real. The onset *time* under every pre-registered Sepsis-3 variant is fixed entirely by the treatment anchor, the organ-dysfunction criterion selecting which stays are labelled without moving any onset by an hour — an implementation choice, not a property of Sepsis-3. Yet a treatment-independent deterioration label agrees with Sepsis-3 at $\kappa = 0.057$, close to chance, so the label effect is not an artefact of re-reading one definition. The treatment anchor alone is as predictable as Sepsis-3 built on top of it (anchor-attributable fraction 1.041, 95% CI 0.990–1.097, including 1), so the organ-dysfunction requirement selects which stays are labelled without selecting the separable ones. Re-timing the onset, by contrast, does change separability: Variant F is 0.0361 AUROC above Variant B on the same patients. Critically, the deterioration label records 1,362 **onsets in the pre-treatment window**, at 0.8498 AUROC: patients do deteriorate before clinicians act, and the empty window is a limitation of how the onset was timed rather than a fact about the patients.

All estimates carry stay-level cluster-bootstrap intervals; the six pre-registered hypotheses are corrected by Holm–Bonferroni and all subgroup analyses by Benjamini–Hochberg. Under correction, subgroup discrimination differences by race are too imprecisely estimated to support any claim, while differential *label* sensitivity does survive: patients whose race is undocumented, and Spanish-speaking patients, change sepsis status across label definitions significantly more often than the reference groups.

The collapse in clinical utility is not a model problem, and on this evidence it is not primarily a discrimination problem either. It is a label problem. Better models cannot fix a label whose *membership* is constituted by the very actions it is meant to anticipate — and how that label’s onset is *timed* determines whether anticipation can be measured at all.

Keywords: sepsis; MIMIC-IV; discrete-time survival analysis; competing risks;

label definition; variance decomposition; clinical utility; temporal dataset shift;
treatment-anchored labels; TRIPOD+AI.

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Chapter 1

Introduction

1.1 Context and Motivation

Sepsis is a life-threatening condition caused by the body’s extreme response to infection [1]. It kills about eleven million people every year, nearly one in five deaths worldwide [2]. In the ICU, every hour of delayed treatment raises the risk of death [3]. An early warning system that detects sepsis before doctors recognise it could save lives by starting treatment sooner.

This potential has driven major research investment. Wang et al. [4] reviewed ninety-one published real-time sepsis prediction models and found a median internal AUROC of 0.811, suggesting strong performance. But the same review found that the clinical Utility Score collapsed from +0.381 internally to −0.164 externally. A negative score means the model’s alerts cause more harm than benefit. AUROC did not detect this collapse because it only measures whether a model can rank patients correctly, not whether its alerts are timely or useful. This gap between apparent accuracy and real-world usefulness has also been observed in deployment: Wong et al. [5] evaluated the Epic Sepsis Model v2 across 227,091 encounters and found high AUROC alongside an alert burden that clinical staff described as unmanageable.

The key question is therefore not “which model is best?” but something more fundamental: *how much of a model’s reported performance comes from the model itself, and how much comes from how the study was designed: how sepsis was defined, how data was split, and which metric was reported?*

Cohen et al. [6] showed on MIMIC-III that varying only the Sepsis-3 definition changed the identified cohort from 867 to 2,178 patients, and that the label choice affected performance as much as the model choice. Kamran et al. [7] showed that published models partly learn to predict when clinicians start treatment rather than when patients deteriorate. Together, these findings suggest that much of the

reported performance may come from the researcher’s design choices, not from the AI model.

This thesis tests that hypothesis directly. Instead of fixing the label, the split, and the metric and asking which model wins, it treats all three as **sources of variance** and measures how much each one contributes to the spread in reported performance.

1.2 Limitations of the Conventional Approach

The standard approach to sepsis prediction follows a simple formula: choose one Sepsis-3 definition, use one train/test split, report AUROC, and compare models. Most of the ninety-one models reviewed by Wang et al. [4] follow this pattern.

This approach has four key problems:

1. **The sepsis label is not fixed.** Cohen et al. [6] showed that different ways of applying Sepsis-3 to the same database can create very different patient groups (867–2,178 cases). The label choice can change results as much as the model choice.
2. **The data split can inflate results.** Guo et al. [8] found that sepsis prediction suffers the largest performance drop of any clinical task when tested on data from a different time period. Random splits let future patterns leak into training, making models look better than they are.
3. **AUROC hides the real failure.** Wang et al. [4] showed that a model can keep a high AUROC while its clinical usefulness drops to harmful levels. The failure is invisible under AUROC alone.
4. **The label captures doctor behaviour.** Kamran et al. [7] showed that models may learn when doctors suspect sepsis and begin treatment, not when the patient’s condition actually worsens.

A study that fixes all of these choices and compares only models is measuring one source of variation while ignoring three others of similar size. This thesis fills in the missing cells of the comparison.

1.3 Research Question and Pre-Registered Hypotheses

Research question. *In real-time sepsis prediction on MIMIC-IV, how much of the reported performance variation comes from (a) the choice of model, and (b)*

the researcher’s decisions: how sepsis is defined, how data is split, when prediction is made, and how performance is measured?

Six hypotheses were pre-registered before any modelling:

- H1 (Label dominance)** Changing the sepsis definition affects AUROC at least as much as changing the model.
- H2 (Coefficient instability)** At least one clinical feature changes its effect direction across label variants.
- H3 (Treatment leakage)** Restricting evaluation to before treatment begins reduces AUROC by at least 0.03.
- H4 (Split optimism)** Time-based testing gives at least 0.02 lower AUROC than random testing.
- H5 (Metric divergence)** The Utility Score declines more than AUROC when moving from internal to external validation.
- H6 (Model-class null)** The primary statistical model performs within 0.02 AUROC of gradient-boosted trees.

H1 and H6 are in deliberate tension: if H1 is confirmed, the label is the main source of variation; if H6 is rejected, the model also matters. Both outcomes are informative.

These six form a single confirmatory family and are tested with Holm–Bonferroni control of the family-wise error rate. Every other analysis in this thesis, including the entire equity analysis, is exploratory and is reported with cluster-bootstrap intervals and Benjamini–Hochberg false-discovery control. Section 3.14 sets out the plan and states plainly that it was written after the analyses had been run.

1.4 Contributions

1. **C1 (Label variance).** To the best of our knowledge, the first three-variant label sensitivity analysis for a dynamic survival model on MIMIC-IV, following Cohen et al. [6] and Dutta et al. [9] (see Section 2.1 for the search strategy supporting this claim).
2. **C2 (Treatment leakage located in the onset rule).** A proof that a conjunction-based Sepsis-3 onset rule — the rule used here and widely in the MIMIC-IV literature — makes pre-treatment prediction structurally impossible, together with the measurement that the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$

rule applied to the identical stays recovers 71 pre-treatment onsets at 0.8081 AUROC. Kamran et al.’s [7] concern is confirmed for label *membership* and narrowed for label *timing*.

3. **C3 (Utility ceiling documented).** Swept across the whole threshold grid rather than scored at a conventional cut-off, no model exceeds 0.039 under the primary label or 0.112 under any pre-registered variant — a few per cent above the strategy of issuing no alerts, on a scale whose upper bound is 1 — and reaching even that costs tens of false alerts per true one. This extends Wang et al.’s [4] meta-analytic finding to a controlled single-study comparison and settles it against the operating-point objection.
4. **C4 (Coefficient instability).** 6 of the 22 shared clinical covariates are unstable across label variants, 2 of them by direction — including vasopressor exposure, a treatment indicator — supporting Lauritsen et al.’s [10] framing-instability finding on MIMIC-IV.
5. **C5 (Equity-by-label analysis).** To the best of our knowledge, the first study to examine both model performance disparities and label sensitivity across demographic subgroups, following Wang et al. [11] (see Section 2.1).
6. **C6 (Reproducible pipeline).** A pre-registered, seed-fixed pipeline following TRIPOD+AI [12] and PROBAST+AI [13] reporting standards.
7. **C7 (Definitional vs. empirical label variance separated).** The Sepsis-3 conjunction is split into its two limbs and each is labelled separately, so that the part of the label effect guaranteed by the definition can be told apart from the part discovered in the data. Two results follow. The onset *time* under every pre-registered Sepsis-3 variant is fixed entirely by the treatment anchor — the organ-dysfunction criterion filters which stays are labelled and moves no onset by a single hour — while a treatment-independent deterioration label agrees with Sepsis-3 at $\kappa = 0.057$, close to chance. And the treatment anchor alone is as predictable as the Sepsis-3 conjunction built on top of it ($\phi_{\text{anchor}} = 1.041$, interval including 1), so the difficulty lies neither in the conjunction nor in the physiology but in when the label is stamped. The same comparison shows the empty pre-treatment window of C2 to be a property of the onset rule, not of the patients: a treatment-independent label records 1,362 onsets there at 0.8498 AUROC, and Sepsis-3 itself records 71 once its onset is timed as the standard specifies (Section 5.6).

1.5 Thesis Structure

Chapter 2 reviews the evidence that motivated this study's design. Chapter 3 describes the data, labels, models, and evaluation plan. Chapter 4 explains the pipeline architecture and its twelve scripts. Chapter 5 presents all results, organised around the six hypotheses. Chapter 6 draws conclusions and outlines future work.

Chapter 2

Literature Review

This chapter summarises the key evidence that shaped the design of this study. A full critical review of fifty sources is provided in the companion document *Statistical and Machine-Learning Approaches to Real-Time Sepsis Onset Prediction on MIMIC-IV: An Extended Critical Literature Review*. This chapter extracts the seven evidence strands that directly informed the study’s design choices. Section 2.1 documents the search strategy used to identify the literature and to support the novelty claims made in this thesis.

2.1 Search Strategy

The literature search was conducted to identify published work on real-time sepsis prediction with a focus on label sensitivity, evaluation methodology, and fairness/equity analysis. Table 2.1 documents the search protocol.

What kind of review this is. This is a **structured narrative critical review**, not a **PRISMA-conformant systematic review**. Searches were executed to a documented protocol and every included source was verified against a primary record, but title-and-abstract screening was not conducted against a full de-duplicated yield, and no second reviewer independently screened. The distinction matters for the novelty claims in Section 1.4: those claims are stated as “to the best of our knowledge” and are bounded by the coverage this method achieves, which is deliberate depth on the methodological literature that constrains this study rather than exhaustive coverage of the MIMIC-IV sepsis corpus. An exhaustive systematic review could surface a prior study combining the two analyses in contribution C5; this search did not.

Element	Detail
Databases searched	PubMed/MEDLINE, Embase, Scopus, Web of Science, IEEE Xplore, ACM Digital Library, Google Scholar, Cochrane Library, ScienceDirect, SpringerLink, Wiley Online Library, Nature Portfolio, JAMA Network, BMJ Journals, arXiv/medRxiv
Searches executed	35 structured searches across the domains listed under “Search terms”
Date range	Primary sepsis-prediction literature: January 2015 – July 2026. Methodological and definitional foundations (e.g. pooled logistic regression, decision-curve analysis) were searched without a date limit, and the earliest source retained dates from 1990
Search terms	(sepsis OR “sepsis prediction” OR “sepsis detection”) AND (MIMIC OR “critical care” OR “intensive care”) combined with each of: “label definition”, “label sensitivity”, “onset definition”, “utility score”, “clinical utility”, “fairness” OR “equity” OR “disparity” OR “bias”, “subgroup analysis”, “competing risks” OR “discrete-time survival”
Inclusion criteria	(1) Peer-reviewed journal article, conference paper, or preprint on a recognised server (arXiv, medRxiv); (2) Develops or evaluates a real-time (hourly or sub-hourly) sepsis prediction model; (3) Uses EHR data from an ICU population; (4) Reports at least one quantitative performance metric
Exclusion criteria	(1) Non-English language; (2) Editorials, commentaries, or letters without original results; (3) Models for sepsis severity staging or mortality only (not onset prediction); (4) Studies on paediatric or neonatal populations exclusively
Verification standard	Every included source was checked against a publisher record, PubMed or PMC record, or institutional repository entry, with journal, volume, year and DOI confirmed. Every numerical claim attributed to a source was located in that source’s own abstract, results text or publisher-hosted record, never inferred from a secondary citation
Total included	50 sources in the companion critical review (46 newly verified in this pass, 4 carried forward from the base review of 19 sources); 33 are cited in this thesis
Excluded at verification	4 items encountered in secondary citation lists could not be verified to the standard above and were excluded; 1 further item is retained with an explicit note of incomplete metadata

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Table 2.1: Literature search strategy. This search supports the novelty claims for contributions C1 and C5: no identified study combines multi-variant label sensitivity analysis with subgroup equity evaluation for a sepsis prediction model. Because the review is narrative rather than systematic, those claims are bounded by the search strategy.

Figure 2.1 summarises the identification, verification and inclusion process. It follows the reporting logic of PRISMA 2020 [14] — an auditable count at every stage — but is presented as a search-and-verification flow rather than as a PRISMA flow diagram, because the screening stage a PRISMA diagram reports was not performed in this form.

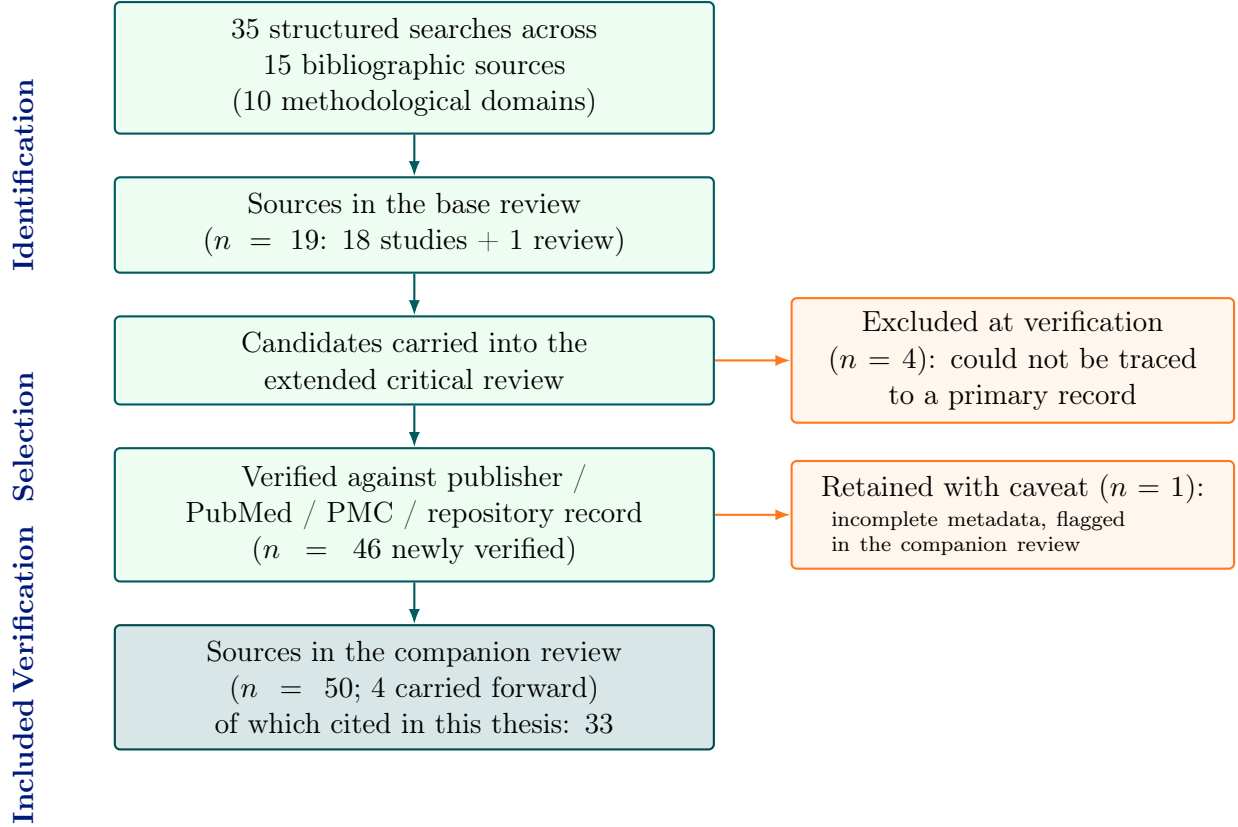


Figure 2.1: Search, verification and inclusion flow. Counts are auditable at every stage, following the reporting logic of PRISMA 2020 [14], but this is not a PRISMA flow diagram: no title-and-abstract screening pass over a full de-duplicated yield was performed, and screening was not duplicated by a second reviewer. The review is narrative and critical, and the novelty claims it supports are bounded accordingly (Section 1.4).

Supporting the novelty claims. Two specific claims require search-based evidence:

1. **C1 (Label variance):** No identified study applies three or more Sepsis-3 label variants to a dynamic survival model on MIMIC-IV. Cohen et al. [6] per-

formed a multi-label analysis on MIMIC-III using tree-based and deep learning models. Dutta et al. [9] tested multiple label definitions prospectively but used a commercial gradient-boosted model, not a survival framework. The present study extends this line of work to a discrete-time competing-risks model on MIMIC-IV v3.1.

2. **C5 (Equity-by-label):** No identified study reports both subgroup model-performance disparities and subgroup label-sensitivity variation for a sepsis prediction model. Wang et al. [11] reported subgroup performance disparities under a single label definition. Studies examining label sensitivity (Cohen et al., Dutta et al.) did not stratify by demographic subgroup. The combination of both analyses is the novel contribution.

Clinical utility assessment criterion. Wang et al. [4] reviewed ninety-one real-time sepsis prediction models. Of these, only a minority reported the PhysioNet Utility Score or an equivalent net-benefit metric; the majority reported only AUROC and/or AUPRC. The claim that published models do not demonstrate good clinical utility is based on the subset of models for which a utility or net-benefit metric was available: among those, the median external Utility Score was -0.164 , indicating net harm. For models reporting only AUROC, clinical utility could not be assessed, and no claim about their utility is made.

Key Concepts

Several domain-specific terms recur throughout this review:

- **MIMIC-IV** (Medical Information Mart for Intensive Care, version IV): a freely available, de-identified database of approximately 74,000 ICU admissions at Beth Israel Deaconess Medical Center, Boston, covering 2008–2019 [15].
- **Sepsis-3:** the current international consensus definition of sepsis as life-threatening organ dysfunction caused by a dysregulated host response to infection, operationalised through a combination of suspected infection (antibiotic order plus culture draw) and a rise in the Sequential Organ Failure Assessment (SOFA) score of ≥ 2 points [1].
- **AUROC** (Area Under the Receiver Operating Characteristic curve): a discrimination metric that measures a model’s ability to rank positive cases above negative cases, ranging from 0.5 (random) to 1.0 (perfect separation). It does not account for alert timing or false-alarm cost.

- **Utility Score:** a net-benefit metric proposed by the PhysioNet/Computing in Cardiology Challenge that rewards early, true alerts and penalises late alerts and false alarms, producing values that can be negative when a model causes more harm than benefit.
- **NEWS2** (National Early Warning Score 2): a standardised clinical scoring system aggregating six physiological parameters (respiratory rate, oxygen saturation, systolic blood pressure, pulse rate, level of consciousness, and temperature) into a single risk score used for bedside deterioration screening.

Research Questions

This literature review is structured around four questions that directly inform the study's design:

- RQ1.** How much does the choice of sepsis label definition affect model performance, and is this effect comparable in magnitude to the choice of model architecture?
- RQ2.** What methodological pitfalls—validation strategy, metric selection, treatment-time leakage—most distort reported performance in the sepsis prediction literature?
- RQ3.** What statistical framework is appropriate for modelling hourly ICU event data when the standard proportional-hazards assumption is violated?
- RQ4.** Does the existing equity literature on sepsis prediction account for label-level variation across demographic subgroups, or only model-level variation?

Current State of the Field

Real-time sepsis prediction has attracted substantial research effort over the past decade. Wang et al. [4] identified ninety-one models published between 2015 and 2024, spanning logistic regression, gradient-boosted trees, recurrent neural networks, and transformer architectures. Despite this volume, most studies share three weaknesses: they use a single label definition, report AUROC as the primary metric, and validate on a random patient-level split. The evidence reviewed in this chapter shows that each of these choices individually inflates or obscures model performance, and their combination produces a literature in which high reported AUROCs coexist with models that fail to improve clinical outcomes at external sites. Table 2.2 provides a structured comparison of the key studies that inform this analysis.

Study	Dataset	Method	Key Finding	Gap Addressed
Johnson et al. [16]	MIMIC-III	5 label algorithms	Agreement between sepsis definitions only moderate ($\alpha = 0.40\text{--}0.62$)	Label ambiguity
Cohen et al. [6]	MIMIC-III	Trees, DL, survival	Label change $\approx$ model change in performance impact	Label as variable
Dutta et al. [9]	9 hospitals (198k)	Commercial GBT	Label sensitivity holds prospectively at scale	External validity
Kamran et al. [7]	MIMIC-IV	Multiple models	AUROC inflated when treatment-time leakage not excluded	Treatment bias
Guo et al. [8]	MIMIC-IV	Domain adaptation	Sepsis worst-affected by temporal shift; random split inflates	Validation strategy
Wang et al. [4]	91-study review	Meta-analysis	AUROC survives external shift; Utility Score collapses	Metric selection
Wang et al. [11]	MIMIC-III	Subgroup analysis	Performance varies by race, ethnicity, insurance, language	Label-by-subgroup gap

Table 2.2: Comparison of key studies reviewed, their datasets, methods, and the design gaps each addresses.

2.2 The Sepsis Label Is Not a Single Object

The common assumption that “the MIMIC-IV sepsis cohort” is one well-defined group is wrong. Two studies show why.

Johnson et al. [16] applied five different sepsis identification algorithms to 11,791 ICU admissions in MIMIC and found only moderate agreement between them (Cronbach’s $\alpha = 0.40\text{--}0.62$). The implication is clear: “the MIMIC sepsis cohort” is not one group but a family of related but different groups, each created by a specific reading of Sepsis-3.

Cohen et al. [6] showed what this means for prediction. On MIMIC-III, using three different but reasonable Sepsis-3 interpretations, the identified cohort ranged

from 867 to 2,178 admissions. Across multiple model types (tree-based, deep learning, and survival models), changing the label definition moved performance by as much as changing the model itself. This is the most important finding behind the present study: it means the label must be treated as a primary variable, not a fixed background choice.

Dutta et al. [9] confirmed this is not a MIMIC-specific artefact. Using a commercial gradient-boosted model across nine hospitals and 198,494 encounters, they found that performance varied substantially depending on whether Sepsis-3, SEP-1, or the CDC Adult Sepsis Event definition was used. The label sensitivity finding holds prospectively, at scale, with a real deployed model.

Figure 2.2 illustrates this core problem: the same patient population, subjected to different but equally valid Sepsis-3 interpretations, produces overlapping but distinct cohorts, each yielding different model performance.

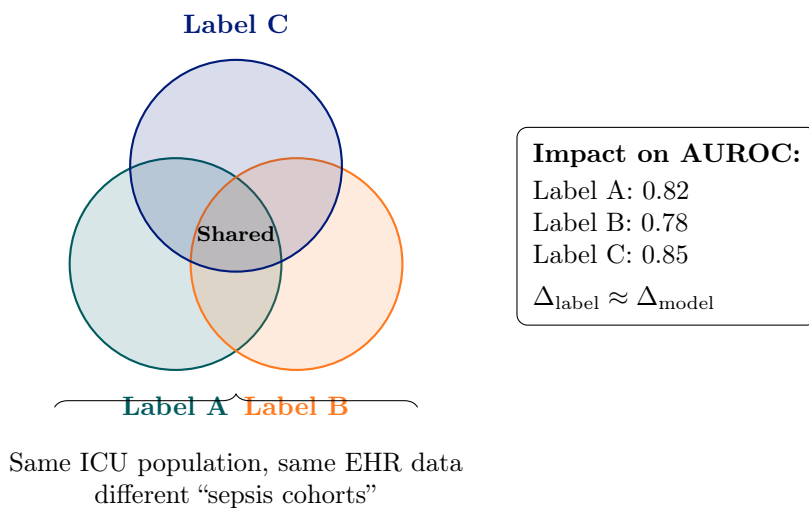


Figure 2.2: Conceptual illustration of label variance: three valid Sepsis-3 interpretations applied to the same ICU population produce overlapping but distinct cohorts, each yielding different model performance. The performance gap between labels is comparable to the gap between model architectures.

2.3 The Label Depends on Clinical Action

The Sepsis-3 label is built from antibiotic orders, culture draws, and SOFA score rises. Kamran et al. [7] observed that clinicians often recognise and begin treating sepsis before the formal criteria are met. A model trained to predict the criteria-meeting time therefore partly learns to predict when the clinician will act. They evaluated performance in the window before treatment started and showed that

apparent AUROC was substantially inflated without that restriction. The model had learned clinician suspicion, not independent physiological deterioration.

Lauritsen et al. [10] formalised the related concept of *framing*: the choice of observation window, prediction horizon, and event trigger. They showed that framing changes can reverse the sign of a feature’s effect; their example is SpO₂. This means a hazard ratio whose sign depends on how the study was framed is not a clinically meaningful quantity.

2.4 The Validation Split Must Be Time-Based

Guo et al. [8] tested four clinical prediction tasks on MIMIC-IV under temporal shift (training on older data, testing on newer data). Sepsis was the worst-affected task. Domain-generalisation and domain-adaptation techniques did not help. A random patient-level split therefore inflates performance more for sepsis than for any other clinical outcome.

This finding drives a key design decision: the primary validation in this study uses a time-based split (train on 2008–2016, test on 2017–2019). A random split is only used to measure how much performance it inflates.

2.5 AUROC Hides the Clinical Failure

Wang et al. [4] reviewed ninety-one real-time sepsis models. Median internal AUROC was 0.811, external AUROC was 0.783, a small, insignificant drop. But median internal Utility Score was +0.381, external was −0.164: the models went from clinically useful to actively harmful. AUROC survived this transition because it only measures whether the model can rank patients by risk. The Utility Score failed because it measures whether alerts arrive in time and at acceptable false-alert rates, and at external sites, they did not. Reporting AUROC as the main metric does not just miss the failure; it actively hides it.

2.6 The Survival Model for Hourly ICU Data

The standard survival approach uses a Cox proportional-hazards model. On MIMIC-IV, the proportional-hazards assumption is rejected (Schoenfeld test, $p < 10^{-16}$). D’Agostino et al. [17] provided the solution: pool observations across short time intervals and use logistic regression for each interval. This pooled logistic regression is mathematically equivalent to time-dependent Cox regression when intervals are short and per-interval event rates are low. One-hour ICU intervals with sepsis onset probability below 1% meet both conditions.

Resche-Rigon et al. [18] and Heyard et al. [19], [20] added the competing-risks extension. In the ICU, patients can leave alive or die without developing sepsis. Treating these events as simple censoring overstates sepsis incidence and biases model estimates. The multinomial cause-specific formulation used in this study handles this correctly.

2.7 NEWS2 as the Baseline Comparator

Most MIMIC-IV sepsis models compare against SIRS, qSOFA, and NEWS. Evans et al. [21] issued a strong recommendation against using qSOFA alone for screening due to poor sensitivity. Mellhammar et al. [22] showed that a LASSO-derived score could not beat NEWS2. The current clinical guidelines have therefore already retired qSOFA as a standalone screening tool, making NEWS2 the minimum meaningful comparator. This study retains SIRS and qSOFA for completeness but uses NEWS2 as the primary rule-based benchmark.

2.8 Equity Analysis: Performance and Label Sensitivity Together

Wang, Li, Naidech and Luo [11] showed that sepsis model performance varies significantly across racial, ethnic, language, and insurance subgroups. However, they did not examine whether the label itself assigns sepsis status differently across those groups. This study adds that missing piece: for each subgroup, it reports both the variation in who gets labelled as septic under each label variant and the variation in model performance. This combination has not, to the best of our knowledge, been reported for any sepsis model (see Section 2.1).

2.9 Summary: Design Decisions and Their Evidence

Table 2.3 maps each major design decision to the evidence that supports it.

Decision	Reason	Source
Three label variants	Label variance $\approx$ model variance	Cohen et al. [6]
Label as variable	Variance is a finding, not a nuisance	Cohen; Dutta et al. [9]
Treatment-anchored eval	Label depends on clinical action	Kamran et al. [7]
Time-based split	Sepsis worst-affected by temporal shift	Guo et al. [8]
Utility Score primary	AUROC hides clinical failure	Wang et al. [4]
Calibration co-primary	Critical for clinical usefulness	Van Calster et al. [23]
Discrete-time model	Cox-equivalent; no PH assumption	D’Agostino [17]; Heyard [19]
NEWS2 comparator	qSOFA no longer recommended	Evans et al. [21]
GBT comparator	Cross-study comparison needs same label	Cohen et al. [6]
Subgroup label sens.	Equity literature omits label-by-subgroup	Wang et al. [11]
Treatment-independent label	Sepsis-3, SEP-1 and CDC ASE all key on treatment timing, so a within-family comparison cannot separate definitional from empirical label variance	Kamran et al. [7]; Dutta et al. [9]

Table 2.3: Design decisions and their evidence warrants.

Chapter 3

Research Methodology

3.1 Research Design

This is a retrospective, observational prediction-modelling study on de-identified critical-care data. It follows a pre-registered protocol (`00_preregistration.md`, locked before any modelling) and reports according to TRIPOD+AI [12] and PROBAST+AI [13] guidelines. The central design innovation is treating the label definition, validation split, and evaluation metric as primary variables rather than fixed background choices.

All code is implemented in R (v4.x) and Python (v3.11), with a single configuration module (`config.R`) as the sole source of truth for all paths, clinical identifiers, and thresholds.

3.2 Data Sources

3.2.1 Development Data: MIMIC-IV v3.1

The primary dataset is MIMIC-IV v3.1 [15], a large de-identified critical-care database from Beth Israel Deaconess Medical Center (BIDMC), Boston. It was accessed under a PhysioNet credentialing agreement. The study uses eight clinical tables (Table 3.1).

Table	Exists	Size (MB)
admissions	yes	19.9
patients	yes	2.8
icustays	yes	3.3
chartevents	yes	3,502.6
labevents	yes	2,593.0
prescriptions	yes	606.7
microbiologyevents	yes	117.2
inputevents	yes	401.3

Table 3.1: MIMIC-IV v3.1 source tables used in this study.

3.2.2 External Validation Data: eICU-CRD v2.0

External validation uses eICU-CRD v2.0 [24], a multi-centre database from over 200 US ICUs. It uses different electronic health record systems and covers a broader range of hospitals and patient populations than the single-centre MIMIC-IV, making it a meaningful external test.

3.3 Ethical Considerations

This study uses two publicly available, de-identified critical-care databases. Neither contains protected health information as defined under HIPAA, and no re-identification was attempted.

MIMIC-IV v3.1 is distributed by the MIT Laboratory for Computational Physiology via PhysioNet [15]. Access requires completion of the Collaborative Institutional Training Initiative (CITI) “Data or Specimens Only Research” course and signing a PhysioNet Credentialed Data Use Agreement (DUA). The DUA prohibits re-identification and restricts data sharing to credentialed collaborators. The principal investigator (the author) completed CITI training and executed the DUA before any data access; credentials are on file with PhysioNet.

eICU-CRD v2.0 [24] is distributed under the same PhysioNet credentialing framework and DUA terms.

Both databases were collected under institutional review board (IRB) approvals held by the original data custodians (Beth Israel Deaconess Medical Center for MIMIC-IV; Philips Healthcare for eICU-CRD), with a waiver of informed consent for the de-identified research dataset. No additional IRB approval was required for secondary analysis of these de-identified records, consistent with the US Common Rule (45 CFR 46.104(d)(4)) and PhysioNet’s data access policies.

3.4 Cohort Construction

The base cohort is built by applying three inclusion criteria uniformly across all label variants:

- Age ≥ 18 at ICU admission.
- First ICU stay per patient only (to prevent within-patient correlation).
- ICU length of stay ≥ 4 hours (to ensure enough data for hourly features and at least one prediction opportunity).

The resulting cohort contains **65,078 stays** in MIMIC-IV. Figure 3.1 shows the patient flow through each exclusion step.

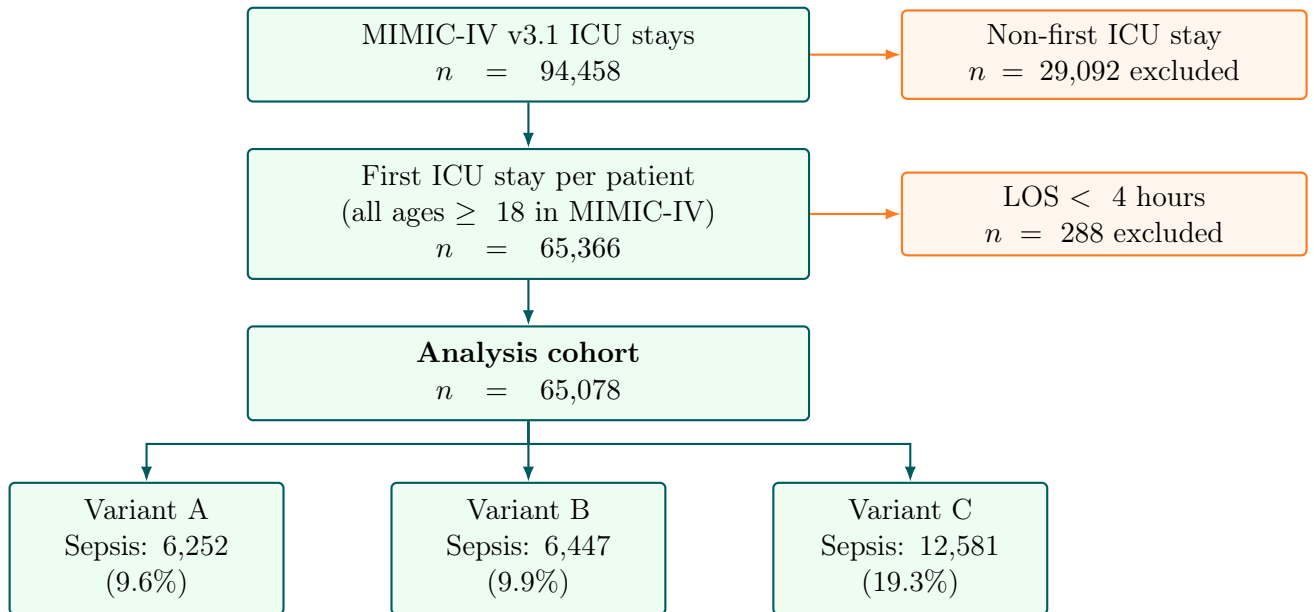


Figure 3.1: CONSORT-style patient flow diagram. All 65,078 stays enter all three label variants; only the sepsis outcome differs. Exclusion counts are from the pipeline’s cohort extraction log (script 02_cohort_labels.Rmd).

3.5 Outcome Definition: Three Sepsis-3 Label Variants

Following Cohen et al. [6] and Johnson et al. [16], the Sepsis-3 onset label is defined three different ways. All three are reasonable readings of the same consensus

definition [1]. They differ in the antibiotic-culture time windows and the SOFA baseline (Table 3.2).

ID	Name	ABX→cx	cx→ABX	SOFA−	SOFA+	Baseline
A	Narrow	24 h	24 h	24 h	12 h	Admission
B	Seymour-Std	72 h	24 h	48 h	24 h	Admission
C	Liberal	72 h	72 h	48 h	24 h	Rolling 24 h

Table 3.2: Three pre-registered Sepsis-3 label variants. ABX = antibiotic; cx = culture; SOFA−/SOFA+ = hours before/after suspicion time in which a SOFA rise of ≥ 2 must be observed. Baseline = SOFA reference value.

In all three variants, a patient has sepsis if there is suspected infection (antibiotic order paired with a culture within the specified windows) and the SOFA score increases by at least 2 points. Surveillance cultures (e.g. MRSA screens) are excluded.

Variant B (Seymour-Standard) is the **primary analysis**. Variants A and C are pre-specified sensitivity analyses for the variance decomposition.

3.6 Beyond Sepsis-3: A Treatment-Independent Label

Variants A, B and C are three readings of one definition, and all three fire on the same trigger: a culture drawn near an antibiotic order. That shared anchor is a record of a clinician’s decision to treat, not a measurement of the patient. Two consequences follow, and they must be separated.

The first is **definitional**. Sepsis-3 [1] makes onset a function of when treatment was given, so any variation in the antibiotic–culture windows necessarily moves the onsets. Part of the label-variant spread reported in Chapter 5 is therefore guaranteed by the construction of the definition and would appear even if the underlying physiology were identical across patients. The second is **empirical**: whether a label that owes nothing to treatment timing picks out different patients, at different hours, with different predictability. Only the second is a claim about the world, and it cannot be settled inside a family of labels that all share the anchor.

Sepsis-3 is a conjunction — *suspected infection* **and** *acute organ dysfunction*. Under Protocol Amendment 2 the conjunction is split and each limb is labelled on its own (Table 3.3).

ID	Name	Suspicion anchor	Organ dysfunction	Uses treatment timing?
B	Seymour-Std	required	required	yes, for membership and timing
E	Anchor-Only	required	not required	yes
D	Deterioration	not required	required	no
F	Sepsis-3 Early-Onset	required	required	membership only

Table 3.3: The two limbs of the Sepsis-3 conjunction, labelled separately (Protocol Amendment 2), and the standard onset timing applied to the intact conjunction (Variant F, Protocol Amendment 4). Variant E is Variant B’s suspicion anchor with the SOFA criterion removed; Variant D is acute organ dysfunction with the infection criterion removed; Variant F has Variant B’s membership exactly and differs only in when the onset is placed.

Variant E (Anchor-Only) takes onset to be the suspicion time itself, using Variant B’s windows, with no organ-dysfunction requirement. Its label is purely a record of clinician behaviour. Predicting it well means predicting who gets cultured and started on antibiotics, which is the quantity a treatment-anchored sepsis label partly encodes. The share of Variant B’s above-chance discrimination that Variant E reproduces,

$$\phi_{\text{anchor}} = \frac{\text{AUROC}_E - 0.5}{\text{AUROC}_B - 0.5}, \quad (3.1)$$

is reported as the **anchor-attributable fraction**.

Variant D (Deterioration) takes onset to be the first hour after hour 1 at which a *physiology-only* SOFA score rises at least 2 points above the patient’s admission baseline (the mean of each component over hours 0–3). "Physiology-only" means the vasopressor and catecholamine terms are withheld, so the cardiovascular component is scored from mean arterial pressure alone. No antibiotic order, culture draw or drug administration enters the definition at any point: Variant D is the treatment-independent labelling that the pre-registered family lacked. Both variants are fitted with the identical model specification, feature set and temporal split, so the label is the only thing that changes.

The PhysioNet/CinC 2019 operationalisation [25] was considered as the *treatment-independent* alternative label and rejected for that purpose. It sets onset to $\min(t_{\text{susp}}, t_{\text{SOFA}})$ with the same culture-plus-antibiotic trigger, so membership is still anchored to treatment timing and it would not have broken the circularity. The same objection applies to SEP-1 and to the CDC Adult Sepsis Event [9], both of which are defined on antibiotic days.

Two limits on interpretation are stated here and carried through the results. Variant E is not a sepsis model and its AUROC is not a sepsis result. Variant D is a threshold function of recorded physiology, and five of the six SOFA components are

themselves model inputs, so its discrimination is partly self-fulfilling and its *level* is not comparable with Variant B's. What is comparable, and what carries the empirical content, is whether Variant D selects different patients at different hours — reported as onset agreement (Cohen's κ) and median onset shift in Section 5.6.

3.6.1 Standard Onset Timing (Variant F, Protocol Amendment 4)

That rejection settles the circularity question and no more. The min rule does not break the anchor — membership still requires the antibiotic-culture pair — but it does change something the pre-registered family holds fixed and never examines: *when* the onset is placed. Anchored *membership* and anchored *timing* are separable properties, and only the second is what the treatment-anchored evaluation window tests.

Variants A–C assign $\text{onset} = t_{\text{susp}}$ whenever the SOFA criterion is satisfied anywhere in the evaluation window, so the SOFA criterion gates membership and contributes no timing information. Seymour et al. [26] and the PhysioNet/CinC 2019 Challenge [25] instead assign

$$t_{\text{onset}} = \min(t_{\text{susp}}, t_{\text{SOFA}}), \quad (3.2)$$

with t_{SOFA} the first hour in the evaluation window at which the ≥ 2 -point rise over the admission baseline is already present. Equation (3.2) is the branch that permits an onset to precede the first antibiotic or culture; dropping it makes the pre-treatment window empty as a matter of arithmetic (Proposition 5.1 in Chapter 5).

Variant F (Sepsis-3 Early-Onset) restores it and changes nothing else. The suspicion windows, SOFA threshold, evaluation window and baseline are Variant B's, so the two variants label exactly the same stays; only the timestamp moves, and only ever earlier. Variant F therefore isolates the timing branch as cleanly as Variants D and E isolate the two limbs.

Three properties are worth stating in advance. First, the SOFA score used for t_{SOFA} is the *full* score, vasopressor components included, as Seymour et al. specify; vasopressor administration is itself a treatment, so an onset placed by the cardiovascular component is free of the antibiotic-and-culture anchor that defines the pre-treatment window but is not free of treatment in general. Variant D remains the fully treatment-independent comparison.

Second, F and B label the same stays septic — membership is identical at the level of the stay — so the informative agreement statistic for F is the onset *shift*, not the overlap. Agreement is nonetheless not perfect once restricted to the modelled horizon, which leads to the third property.

Third, moving an onset earlier can move it *into* the 72-hour prediction window. A stay whose t_{susp} falls after hour 72 contributes no labelled event to any model here, but if its SOFA rise occurred inside the window, Variant F places the onset there and the stay becomes a modellable event. This is a one-directional effect: $F \supseteq B$ within the horizon, and the extra stays are reported in Section 5.6.3. It means the choice of onset rule determines not only *when* events are recorded but *how many* are visible to a real-time model at all.

Variant F is post-hoc, is never pooled with A–C, and takes no part in H1 or the variance decomposition. Its contrast against B joins exploratory family E3, which enlarges that family and makes every Benjamini–Hochberg adjustment within it stricter than before the amendment.

Variants D, E and F are post-hoc and are never pooled with A–C. They take no part in H1 or in the variance decomposition table, which are computed from exactly the same inputs as before the amendments; their contrasts form their own exploratory family (E3), corrected by Benjamini–Hochberg.

3.7 Prediction Framing

Following Lauritsen et al. [10] and Reyna et al. [25]:

- **Observation window:** Expanding from ICU admission, uses all data available up to the current hour.
- **Prediction horizon:** 6 hours (will sepsis onset occur within the next 6 hours?).
- **Prediction frequency:** Every hour.
- **Start time:** 1 hour after ICU admission.
- **End time:** 72 hours after ICU admission (right-censored).

This framing matches the PhysioNet/CinC 2019 Challenge [25].

3.8 Person-Hour Table and Competing Events

Each patient stay is expanded into a table with one row per person-hour at risk. Following Heyard et al. [19], each hour is assigned one of four outcomes:

- 0: Still at risk (censored at this hour)
- 1: Sepsis onset (the event of interest)

- 2: Discharged alive without sepsis (competing event)
- 3: Died without sepsis (competing event)

Treating discharge and death as simple censoring would overstate sepsis incidence and bias model estimates. The multinomial competing-risks formulation avoids this problem.

3.9 Feature Construction

Features are extracted at each person-hour from chartevents, labevents, inpuvents, and outpuvents. The feature set includes:

1. **Vital signs** (last value carried forward per hour): heart rate, respiratory rate, SpO₂, MAP (invasive or noninvasive), temperature (°C), GCS total.
2. **Laboratory values** (last value carried forward): lactate, creatinine, total bilirubin, platelet count, WBC, P/F ratio.
3. **Vasopressor indicators**: any vasopressor (yes/no); any norepinephrine or epinephrine (yes/no).
4. **Six-hour trends**: rolling slopes for respiratory rate, MAP, and lactate.
5. **Missingness indicators**: whether each lab test was ordered in the current hour (one flag per test).

Temperature in Fahrenheit is converted to Celsius. FiO₂ values above 1.0 (percentage format) are divided by 100. In total, 25 candidate predictors are used.

3.10 Model Classes

3.10.1 Primary Model: Dynamic Discrete-Time Competing-Risks Hazard Model

The primary model is a multinomial pooled logistic regression over person-hours, following D’Agostino et al. [17] and Heyard et al. [19]:

$$\lambda_r(t \mid \mathbf{X}_{it}) = \frac{\exp(\alpha_r(t) + \boldsymbol{\beta}_r^\top \mathbf{X}_{it})}{1 + \sum_s \exp(\alpha_s(t) + \boldsymbol{\beta}_s^\top \mathbf{X}_{it})} \quad (3.3)$$

where $r \in \{1 = \text{sepsis}, 2 = \text{discharge}, 3 = \text{death}\}$. The baseline hazard $\alpha_r(t)$ uses a restricted natural cubic spline with 5 knots, placed at quantiles of training-set hour within each variant.

This model predicts the risk of sepsis every hour. Under short intervals and low per-interval event probabilities (both conditions met for hourly ICU data with onset probability below 1%), it is mathematically equivalent to time-dependent Cox regression, without requiring the proportional-hazards assumption.

Identification: the specification is penalised, and must be. The unpenalised version of Equation (3.3) is **completely separated** on this design. Driven to a tight tolerance by iteratively reweighted least squares, the whole coefficient vector diverges together — $|\beta|$ of order 10^{15} under *every* label variant — which is the signature of separation along a direction in covariate space rather than of one badly behaved predictor; the design matrix is full rank, so this is not aliasing. A separated likelihood has no finite maximiser, so the fitted coefficients are not identified and the point at which a quasi-Newton optimiser stops is arbitrary.

This is a trap worth naming, because the standard diagnostic does not catch it: `nnet::multinom` returns `convergence = 0` for every variant here. That flag reports only that BFGS’s relative-change stopping rule fired; it says nothing about whether an optimum exists. A ridge penalty of $\lambda = 10^{-4}$ is therefore applied, which makes the penalised log-likelihood strictly concave and the maximiser unique. The value is **not tuned**: across a 20-point ridge path the temporal-test AUROC moves by less than 0.02 for every variant, and that insensitivity — not an optimisation — is the justification for the constant. Every fit is checked after estimation and flagged if $\max |\beta|$ exceeds 20.

Penalisation is not a refinement here; it is a precondition. Coefficient comparisons across label variants (H2, Section 5.5.4) are only meaningful against a unique maximiser, and an unpenalised fit can land in a degenerate region under one label and a benign region under another, manufacturing an apparent label effect where none exists.

Uncertainty for this model comes from the bootstrap, not from the fit. A cluster-robust sandwich estimator clustered on stay would be the natural choice for standard errors, since person-hours are nested within stays. It is not available: `sandwich`’s score-function method is not defined for `multinom` objects, so no analytic variance is computed for the primary model at all. All uncertainty reported for it therefore comes from the stay-level cluster bootstrap of Section 3.14, which resamples whole stays and so respects the same clustering the sandwich estimator would have. The one quantity this cannot supply is a per-coefficient standard error, which is why H2 is reported as a descriptive count rather than a

test (Section 5.5.4).

3.10.2 Comparator: Gradient-Boosted Trees (GBT)

An XGBoost model is fitted under the same conditions for each label variant and split. Settings: maximum depth 6, learning rate 0.1, early stopping on a 15% internal validation set, log-loss objective.

3.10.3 Comparator: Static Cox Model

A standard Cox proportional-hazards model using only the first observation per stay, included as a deliberately weak baseline to show the cost of ignoring time-varying data. Schoenfeld residual testing is applied to verify the expected violation of proportional hazards.

3.10.4 Rule-Based Comparators

NEWS2, qSOFA, and SIRS are computed at each person-hour from the same features. NEWS2 is the headline comparator following Evans et al. [21].

3.11 Validation Architecture

Split	How	Why
Internal (primary)	Time-based: train 2008–2016, test 2017–2019 [8]	Honest internal estimate
Internal (secondary)	Random 75/25 patient-level split (seed = 42)	Measure split optimism (H4)
Treatment-anchored	Hours before first culture or antibiotic [7]	Treatment leakage (H3)
External	eICU-CRD v2.0 [24], no retraining	Generalisability (H5)

Table 3.4: Validation architecture. All splits are patient-level: no patient appears in both training and test sets.

3.12 Evaluation Metrics

3.12.1 Primary Metrics

PhysioNet Utility Score [25] is the primary metric. It rewards early true alerts and penalises late and false ones. AUROC is reported only for comparison, because

it hides the clinical failure mode [4].

Each stay contributes a raw utility. A septic stay first alerted at hour t within the reward window $[t_{\text{onset}} - 6, t_{\text{onset}}]$ earns $(t_{\text{onset}} - t)/6$, so a full-horizon warning earns 1 and an alert that merely coincides with onset earns nothing; a septic stay alerted only in $(t_{\text{onset}}, t_{\text{onset}} + 3]$ earns -0.05 per late alert hour; a septic stay never alerted in either window earns -1 ; and every alert hour outside a reward window, on any stay, costs -0.05 .

The raw total U is then referenced to the two strategies that bound it. The *inaction* strategy never alerts and therefore earns $U_{\text{inact}} = -1$ per septic stay; the *optimal* strategy earns the full reward on every septic stay and raises no false alarm, giving $U_{\text{opt}} = +1$ per septic stay. The reported score is

$$\tilde{U} = \frac{U - U_{\text{inact}}}{U_{\text{opt}} - U_{\text{inact}}}, \quad (3.4)$$

so that $\tilde{U} = 1$ is a perfect forecaster, $\tilde{U} = 0$ is *exactly* the no-alert strategy, and $\tilde{U} < 0$ is worse than issuing no alerts at all. The inaction baseline has to appear in both numerator and denominator: normalising by the optimum alone would score a model that raises no alert at -1 rather than at 0 , contradicting the meaning of the metric. Equation (3.4) is verified against both bounds at run time — a threshold above every predicted probability is required to return exactly zero, and no operating point may exceed one.

Operating points. A score of this kind is a property of a model *and* a threshold. At an hourly event rate below 0.5 per cent no model’s predicted probability approaches the conventional 0.5 cut-off, so that cut-off measures the no-alert strategy for a well-calibrated model and an arbitrary point for a poorly calibrated one. The Utility Score is therefore reported twice: at the 0.5 cut-off, for comparability with the literature, and at its maximum over a threshold grid. The grid is defined on the *alert rate* — the fraction of person-hours alerted — rather than on the probability scale, spanning 10^{-5} to 0.5 over sixty logarithmically spaced points, so that it adapts to each model’s calibration and covers every operating point a ward could contemplate. The maximum is taken over thresholds that raise at least one alert, and the alert burden, alert rate and per-stay detection rate that attain it are reported alongside, because a utility maximum bought at an unusable alarm volume is not a deployable operating point.

Calibration is assessed via calibration intercept and slope, following Heyard et al. [20], and, following Van Calster et al. [23], is additionally reported as a flexible calibration curve (restricted cubic spline of the outcome on the logit of predicted risk) summarised by the Integrated Calibration Index (ICI), E50, E90, E_{max}, and the scaled Brier score. Predicted probabilities are floored at 10^{-6} rather than at the conventional 10^{-3} , because at an hourly event rate below 0.5 per cent the conventional guard overwrites the majority of the distribution being measured.

3.12.2 Secondary Metrics

AUPRC (important at low event rates), alert burden (false alerts per true alert; benchmark: 1.4 from Moor et al. [27]), lead time (benchmark: 3.7 hours from Moor et al.), and decision-curve net benefit [28]. Net benefit is evaluated on a threshold grid running from one tenth to ten times the observed event prevalence rather than on the conventional 0.01–0.20 grid, which at this prevalence lies entirely above the range of thresholds any deployment would use.

3.12.3 Deployment-Facing Metrics

AUROC, AUPRC and the Utility Score characterise a model but not the alert load a ward would carry. The following are therefore reported alongside them, at a threshold set to the largest cut-off achieving a fixed per-hour sensitivity of 80 per cent rather than at an arbitrary 0.5 cut-off:

- **Positive predictive value** and its reciprocal, the number needed to evaluate (NNE), at that fixed sensitivity.
- **False alerts per patient-day**, which normalises alarm volume by exposure time rather than by the number of true alerts.
- **Per-patient (event-level) detection**: the fraction of septic stays alerted within the actionable window [onset – 6 h, onset], the fraction alerted at any point before onset, patient-level PPV, and the fraction of all stays alerted. Per-hour and per-patient views of the same operating point are reported side by side, because the choice of denominator moves the headline PPV by more than an order of magnitude.
- **False-alarm episodes per patient-day**, counting a contiguous run of alert hours once rather than once per hour.
- **Lead time** from first alert to onset: median, interquartile range, and the fraction of septic stays alerted at least 3 and 6 hours ahead.

The same quantities are computed on eICU-CRD with thresholds re-derived on the external cohort.

3.13 Sample Size

Sample size is calculated using the Riley et al. [29] method, targeting a shrinkage factor ≥ 0.9 with 25 predictors and an anticipated AUC of 0.846 from Moor et al. [27]. Each label variant is evaluated separately; an underpowered variant is a finding, not a failure.

3.14 Statistical Inference and Multiplicity

3.14.1 Uncertainty: the Stay-Level Cluster Bootstrap

Person-hours are nested within ICU stays: a single stay contributes on average about 45 highly correlated hourly rows. Treating those rows as independent would produce standard errors several times too small. Every interval in this thesis therefore comes from a **stay-level cluster bootstrap**, in which whole ICU stays are drawn with replacement and all person-hours belonging to a drawn stay are carried with it. Intervals are 95% percentile intervals from $B = 2,000$ replicates for internal and subgroup analyses and $B = 1,000$ for eICU-CRD, which contains 7.9 million person-hours. The seed is fixed at 42.

Contrasts between quantities measured on the *same* person-hours are evaluated within replicate, so the interval accounts for their correlation. This applies to the label-variant comparison (all three variants are built on the same test stays) and to the model-class comparison (all models score the same hours). Contrasts across *different* evaluation sets — temporal versus random split, and MIMIC-IV versus eICU-CRD — use independent bootstrap streams and propagate both variances.

For subgroup analyses the bootstrap is stratified: stays are resampled within each subgroup so the observed subgroup sizes are held fixed. The subgroups are the object of study, not a random draw from a larger frame.

P-values are obtained by recentring the bootstrap distribution on the null boundary and reading off the tail area, using the $(1 + \text{count})/(B + 1)$ convention so that no p-value is reported as exactly zero. The smallest reportable p-value is therefore $1/(B + 1)$.

3.14.2 Confirmatory and Exploratory Families

The six pre-registered hypotheses form a single **confirmatory** family and are corrected by Holm–Bonferroni at a family-wise error rate of 0.05. The family size is fixed at $m = 6$: the number of hypotheses that were pre-registered, not the number that turned out to be testable. H2 and H3 each reserve a slot without contributing a p-value (Section 5.5 explains why neither admits one), which is deliberately conservative. A hypothesis becoming untestable after the data were seen must not be allowed to relax the correction applied to the others.

All subgroup analyses are **exploratory**. The pre-registration fixed their *scope* — that race, language and insurance would be examined — but not the subgroup levels, the reference level, or the direction of any comparison. They are corrected by Benjamini–Hochberg at $q = 0.05$ within three families: E1, all subgroup AU-ROC contrasts against the reference level across every equity variable and both models; E2, all subgroup label-sensitivity contrasts; and E3, the alternative-label

contrasts of Protocol Amendment 2, in which the AUROC under each limb of the Sepsis-3 conjunction (Section 3.6) is compared with Variant B for both models. The anchor-attributable fraction of Equation 3.1 is a ratio with no meaningful null at zero and is reported as an interval only, outside the corrected family. The correction denominator is the number of contrasts *computed*, not the smaller number selected for presentation in Chapter 5. Bonferroni-adjusted p-values are also reported so that a reader preferring family-wise control over false-discovery control can substitute them.

The two procedures differ by design. A confirmatory claim must survive family-wise control, because one false confirmation invalidates the claim. An exploratory scan exists to generate hypotheses and needs only its false-discovery proportion bounded; applying family-wise control to it would suppress every signal the scan is meant to surface.

Each subgroup is contrasted against the largest level of its own variable. The reference is chosen by sample size alone, never by outcome. A subgroup is reported only if it has at least 50 person-hours and at least 5 sepsis onsets; below that floor an AUROC is not estimable and the subgroup is listed as such rather than shown as a point estimate.

3.14.3 Analysis Status Tagging

Every result in Chapter 5 carries an explicit status tag rather than relying on a single blanket statement in this chapter:

- **Confirmatory** — estimand, threshold and direction were fixed in `00_preregistration.md` before any modelling.
- **Exploratory** — pre-registered in scope but not in specific contrasts. Generates hypotheses; does not test them.
- **Descriptive** — no inferential claim is attached.

The tags are held in a machine-readable analysis register (`12_analysis_register.parquet`) produced by the pipeline, and are reproduced in Table 5.27.

3.14.4 Status of the Inference Plan Itself

This inference and multiplicity plan is **Protocol Amendment 1**, dated 30 July 2026 and recorded as §13–15 of `00_preregistration.md`. It was written *after* the analyses of Chapter 5 had been run and inspected, and it is therefore itself post-hoc. It is reported as such rather than presented as pre-specification.

The amendment changes no hypothesis, estimand, threshold, direction, subgroup or model. It attaches uncertainty to contrasts that were already pre-specified and imposes the multiplicity control the original protocol omitted. Where the correction changes a verdict recorded in the original protocol, the corrected verdict is the one reported, and the change is stated explicitly.

3.15 Pre-Registration and Reporting

The full protocol was locked before any data extraction. All six hypotheses, label variants, model classes, splits, metrics, and decision thresholds are specified in `00_preregistration.md`. Any analysis not specified there is labelled as exploratory, and, following Section 3.14, each individual result carries its status tag in the results table itself rather than inheriting it from this statement alone. Reporting follows TRIPOD+AI [12]; code availability and the constraints the data use agreement places on it are set out in Section 4.14.2.

Chapter 4

System Design and Pipeline Architecture

The pipeline consists of twelve numbered, single-purpose scripts in `v2-proj/`, preceded by the locked pre-registration document (`00_preregistration.md`). Each script reads from upstream outputs and writes its own results to `v2-proj/output/`. No script modifies the original source data. All random seeds are fixed for reproducibility. Figure 4.1 shows the data flow.

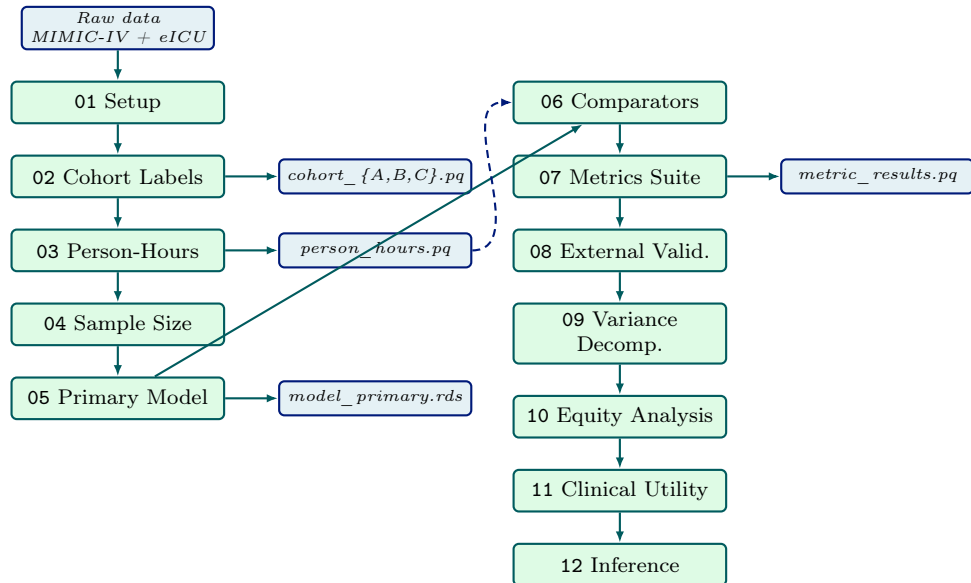


Figure 4.1: Pipeline data flow. Each box is a single script; italic nodes are output files. Raw MIMIC-IV and eICU data are read-only.

4.1 Configuration and Utilities

`config.R` is the single source of truth for all pipeline settings: data paths, cohort criteria, antibiotic name lists, MIMIC-IV item identifiers, label variant definitions, split boundaries, prediction framing constants, equity variable names, and Utility Score parameters. No other script hard-codes a clinical constant or file path.

`utils.R` provides shared helper functions: a DuckDB-backed streaming reader for large CSVs, Parquet reader/writer wrappers, datetime parsing, last-observation-carried-forward (LOCF) imputation, and Schoenfeld test wrappers.

The full pipeline can be reproduced on any MIMIC-IV installation by setting a single environment variable (`MIMIC_RESEARCH_ROOT`).

4.2 Stage 01: Setup Validation

`01_setup.Rmd` checks that all required MIMIC-IV source tables exist and reports their sizes. The pipeline stops if any table is missing.

4.3 Stage 02: Cohort Extraction and Labelling

`02_cohort_labels.Rmd` builds all cohorts in a single pass:

1. **Base cohort:** Adult first ICU stays with $\text{LOS} \geq 4$ h (65,078 stays). Equity variables (race, language, insurance) are attached.
2. **Suspected infection:** Qualifying antibiotic-culture pairs within each variant's time windows. Surveillance cultures are excluded.
3. **SOFA organ dysfunction:** SOFA components are computed from `chartevents`, `labevents`, `inpuvents`, and `outpuvents`. A SOFA rise ≥ 2 within each variant's windows defines organ dysfunction.
4. **Onset time:** For the pre-registered variants A–C, the suspicion time t_{susp} , provided the SOFA rise occurs somewhere in the evaluation window. The SOFA criterion therefore gates *membership* and contributes no timing information — an implementation choice with consequences developed in Section 3.6.1.
5. **Alternative labels (Protocol Amendments 2 and 4):** The Sepsis-3 conjunction is also split into its two limbs (Section 3.6). Variant E keeps the suspicion anchor and drops the SOFA criterion; Variant D keeps organ dysfunction and drops the anchor, so its onsets are derived from physiology in

Stage 03 rather than here. Variant F (Amendment 4) keeps the conjunction intact and re-times the onset to $\min(t_{\text{susp}}, t_{\text{SOFA}})$, also in Stage 03 because it needs the hourly SOFA trajectory. All three are held outside the pre-registered variant set in `config.R` and their outputs are written to separate files so they cannot enter H1 or the decomposition table.

Outputs: `cohort_A/B/C.parquet`, `label_variance_table.parquet`, `cohort_D/E.parquet`, and `alt_label_variance_table.parquet`.

A single environment variable, `V2_VARIANTS`, restricts the per-variant loops in Stages 02–07 to a named subset. Adding the alternative labels therefore did not require refitting variants A–C: XGBoost is not bit-stable across runs, so an incidental refit would have moved every GBT-dependent number in this thesis without any change of substance.

4.4 Stage 03: Person-Hour Table Construction

`03_person_hours.Rmd` expands each stay into hourly rows. Large source tables (chartevents: 3.5 GB; labevents: 2.6 GB) are processed via DuckDB streaming, pre-filtered to cohort stays, to avoid loading everything into memory. For each person-hour:

- Vitals and labs are LOCF-imputed from the most recent measurement.
- Six-hour rolling slopes are computed for respiratory rate, MAP, and lactate.
- Vasopressor indicators are set from concurrent infusions.
- Missingness flags are set for each lab test.
- The competing-event code (0/1/2/3) is assigned based on the hour’s relationship to onset, discharge, or death.
- The temporal split group is assigned from the stay’s admission year.
- Treatment anchor timestamps (first antibiotic, first culture) are recorded.

Outputs: `person_hours_A/B/C.parquet`.

4.5 Stage 04: Sample Size Calculation

`04_sample_size.Rmd` applies the Riley et al. [29] criteria to the training portion of each person-hour table (2008–2016). Sufficiency is evaluated per label variant.

4.6 Stage 05: Primary Model

`05_primary_model.Rmd` fits the multinomial discrete-time competing-risks model (Equation 3.3) for each label variant on the temporal training split. Baseline hazard splines use 5 knots. Standard errors are cluster-robust (grouped by stay). A random patient-level 75/25 split is also fitted to support H4.

Outputs: `model_primary_A/B/C.rds`, coefficient tables, and `predictions_primary_A/B/C.parquet`

4.7 Stage 06: Comparator Models

`06_comparators.Rmd` fits the XGBoost GBT model, the static Cox model (with Schoenfeld testing), and computes NEWS2, qSOFA, and SIRS at each person-hour. GBT models are fitted under both temporal and random splits. GBT feature importance is exported for each variant.

Outputs: model objects, predictions, feature importance tables, and Schoenfeld test results.

4.8 Stage 07: Metrics Suite

`07_metrics_suite.Rmd` evaluates all models on all splits (temporal, random, treatment-anchored) using the full metric set: AUROC, AUPRC, calibration intercept and slope, PhysioNet Utility Score, alert burden, lead time, and decision-curve net benefit.

The Utility Score is computed at the default 0.5 cut-off and swept across a sixty-point grid of alert rates, in a single pass: the per-stay reward is monotone in a falling threshold, so the whole grid costs one sort plus one traversal of the reward-window rows. Two invariants are asserted at run time rather than assumed — a threshold above every predicted probability must return exactly zero, since it *is* the no-alert strategy, and no operating point may exceed the optimal strategy's score of one. A stand-alone Python implementation of the same scoring rule, `utility_score.py`, is kept in step with the notebook so that prediction tables outside the pipeline can be scored and the two implementations cross-checked.

Output: `07_metric_results.parquet` (and `07_metric_results_random.parquet` for the random-split arm), `07_utility_sweep.csv` for the full threshold grid and `07_utility_max.csv` for the attaining operating points.

4.9 Stage 08: External Validation on eICU-CRD

`08_external_validation.Rmd` applies the frozen MIMIC-IV models (primary and NEWS2) to eICU-CRD without any retraining. AUROC and AUPRC are reported for each variant and model.

4.10 Stage 09: Variance Decomposition

`09_variance_decomposition.Rmd` computes the pre-registered variance decomposition by varying one factor at a time while holding all others fixed:

- **Label spread (H1):** AUROC range across Variants A/B/C.
- **Model-class spread:** AUROC range across all models.
- **Split optimism (H4):** Random vs. temporal split comparison.
- **Metric divergence (H5):** AUROC vs. Utility Score decline.

It then runs the **label-anchor attribution** of Protocol Amendment 2: performance under each limb of the Sepsis-3 conjunction, the anchor-attributable fraction of Equation 3.1, and onset agreement (Cohen's κ , median onset shift) between every label variant and Variant B.

Outputs: decomposition tables, hypothesis verdicts, comparison plots, `label_anchor_attribution` and `label_agreement`.

4.11 Stage 10: Equity Analysis

`10_equity_analysis.Rmd` evaluates model performance and label sensitivity within each stratum of the equity variables declared in `config.R` (race, ethnicity, language, insurance). MIMIC-IV v3.1 records race and ethnicity in a single field, so three variables carry data and are reported: **race, language, and insurance**. Levels are restricted to the eight largest per variable. The combination of performance disparities and label-sensitivity variation is the novel finding.

4.12 Stage 11: Clinical Utility Metrics

`11_clinical_metrics.Rmd` computes flexible calibration curves, decision-curve net benefit, operating points at a fixed per-hour sensitivity, per-patient (event-level) detection and alarm-episode counts, and lead time to onset, for both MIMIC-IV and eICU-CRD.

4.13 Stage 12: Inference and Multiplicity

`12_inference.Rmd` is the inference layer. It reads only the Parquet outputs of the earlier stages, so it can be re-run without refitting anything, and it produces three things: stay-level cluster-bootstrap confidence intervals for every headline and subgroup estimate; the Holm-corrected confirmatory tests of H1–H6 and the Benjamini–Hochberg-corrected exploratory families (E1 and E2 subgroup analyses, E3 alternative-label contrasts); and the machine-readable analysis register that supplies the status tags used throughout Chapter 5. The bootstrap resamples ICU stays, never person-hours, and evaluates AUROC and AUPRC as weighted rank statistics on pre-sorted data so that 2,000 replicates over half a million person-hours remain tractable. Setting `V2_BOOT_B` overrides the replicate count for a fast smoke test; reported results always use the values fixed in `config.R`. The statistical rationale is given in Section 3.14.

4.14 Reproducibility

The pipeline is orchestrated by `run_pipeline.R` (or the equivalent `run_pipeline.sh`), which executes all stages sequentially or by phase, renders each stage to PDF in `output/pdf/`, and writes a per-stage and a combined run log to `output/logs/`. All intermediate outputs are Parquet (compressed, columnar). Small result tables (<1,000 rows) are additionally written as CSV for readability. The end-to-end run reported in this thesis completed in 60 minutes on an Apple silicon workstation; the dominant cost is stage 05, which fits the multinomial model on 2.3 million person-hours for each label variant.

4.14.1 Fixed Random Seed

All random operations use a single fixed seed (42) to ensure bitwise reproducibility. Every Rmd script sets `set.seed(42)` in its setup chunk before any computation. The same value is used for the patient-level random split (`RANDOM_SPLIT_SEED` in `config.R`), the `pmsamplesize` sample-size calculation, and all model-fitting calls. No seed is drawn from wall-clock time or left to the system default.

The full pipeline can be reproduced on any MIMIC-IV v3.1 installation by setting the environment variable `MIMIC_RESEARCH_ROOT` to the directory containing the `data/` folder.

4.14.2 Code Availability

The complete pipeline — the twelve numbered scripts, `config.R`, `utils.R`, `utility_score.py`, the pre-registration document and the orchestration scripts — is held in a version-controlled repository and is available from the author on request, including to examiners. The repository is currently private; the licence under which it will be released has not yet been fixed and is stated here as undetermined rather than asserted.

The repository contains **code only**. No MIMIC-IV or eICU-CRD record, derived person-hour table, or fitted model object is included: the `data/`, `processed_data/` and `output/` directories are excluded from version control by design, because the PhysioNet Credentialed Data Use Agreement (Section 3.3) prohibits redistribution of the underlying data. A reader wishing to reproduce these results must obtain MIMIC-IV v3.1 and eICU-CRD v2.0 through their own PhysioNet credentialing, then re-run the pipeline; with the fixed seed above it reproduces the reported estimates exactly.

Chapter 5

Results

This chapter presents the results in eleven sections: cohort description, sample size adequacy, model performance and calibration, clinical utility (net benefit, operating points, event-level detection and lead time), variance decomposition and the six hypothesis verdicts, external validation, proportional-hazards testing, GBT feature importance, equity analysis, the analysis register, and a closing discussion.

Every result below carries an explicit status tag — **[C]** confirmatory, **[E]** exploratory, **[D]** descriptive — defined in Section 3.14 and enumerated in full in Table 5.27. Confirmatory results test a hypothesis whose estimand, threshold and direction were fixed before any modelling; exploratory results were pre-registered in scope only and generate hypotheses rather than testing them. All intervals are 95% percentile intervals from a stay-level cluster bootstrap. The six pre-registered hypotheses are corrected by Holm–Bonferroni at a family-wise error rate of 0.05; the exploratory subgroup families are corrected by Benjamini–Hochberg at $q = 0.05$.

5.1 Cohort Description and Label Variance **[D]**

Table 5.1 summarises the three cohorts. The base extraction identifies 65,078 adult first ICU stays. The database and patients are the same across all three variants; only the sepsis definition differs.

Var	Name	Stays	Onsets	%	Deaths	Onset h (IQR)
A	Narrow	65,078	6,252	9.6	5,934	10[4–26]
B	Seymour-Std	65,078	6,447	9.9	5,892	11[4–30]
C	Liberal	65,078	12,581	19.3	4,956	10[4–30]

Table 5.1: Cohort description by label variant. The Liberal variant identifies about twice as many sepsis cases as the Narrow or Seymour-Standard variants, mainly because it accepts wider antibiotic-culture windows and uses a rolling SOFA baseline.

The twofold difference in sepsis prevalence between Variant A (9.6%) and Variant C (19.3%) confirms the label-sensitivity finding from Johnson et al. [16] at MIMIC-IV scale.

5.2 Sample Size Adequacy [C]

Var	Name	Train h	Events	EPH rate	Min-N	EPV
A	Narrow	2,304,923	4,647	0.00202	52,201	185.9
B	Seymour-Std	2,301,842	4,847	0.00211	51,012	193.9
C	Liberal	2,120,376	9,403	0.00443	24,069	376.1

Table 5.2: Sample size results (Riley et al., 2019). EPV: events per candidate predictor variable (25 predictors). All three variants exceed the minimum required (52,946 stays each). Sample size is not the bottleneck.

5.3 Model Performance: Internal Temporal Evaluation [C]

Table 5.3 reports performance on the temporal test set (2017–2019) under the unrestricted evaluation window.

Var	Model	AUROC	AUPRC	Utility (0.5)	Utility (best)	Alert burden
A	Primary	0.7713	0.0093	0.000	0.026	∞
A	GBT	0.7448	0.0093	-1.699	0.040	195.6
A	NEWS2	0.6413	0.0032	-0.509	-0.001	236.4
A	Cox	0.6589	0.0035	-4.790	0.000	356.4
A	qSOFA	0.6012	0.0027	-2.293	-0.263	343.1
A	SIRS	0.6157	0.0032	-3.209	-0.035	346.9
B	Primary	0.7590	0.0096	0.000	0.025	∞
B	GBT	0.7460	0.0090	-1.843	0.039	194.7
B	NEWS2	0.6415	0.0032	-0.491	0.000	246.7
B	Cox	0.6444	0.0033	-4.720	0.000	361.6
B	qSOFA	0.6091	0.0028	-2.193	-0.252	316.8
B	SIRS	0.6114	0.0030	-3.086	-0.035	351.7
C	Primary	0.7695	0.0201	0.000	0.095	∞
C	GBT	0.7559	0.0191	-0.603	0.112	90.8
C	NEWS2	0.6483	0.0074	-0.099	0.001	102.2
C	Cox	0.6733	0.0085	-1.613	0.003	148.7
C	qSOFA	0.6012	0.0060	-0.705	-0.048	150.9
C	SIRS	0.6242	0.0070	-1.065	0.000	148.4

Table 5.3: Internal temporal test performance (2017–2019, all variants use temporal split). N test rows: 548,827 (B). **Utility (0.5)** is the normalised PhysioNet Utility Score at the default cut-off; **Utility (best)** is its maximum over the threshold grid of Section 3.12. Zero is the no-alert strategy under both. Alert burden is at the 0.5 cut-off (∞ = no true alerts there); the benchmark is 1.4 [27]. Per-threshold curves are in Figure 5.1 and the attaining operating points in Table 5.5. Confidence intervals for Variant B are in Table 5.4. **[C] confirmatory.**

Model	AUROC	95% CI	AUPRC (95% CI)
Primary	0.7590	0.7434–0.7749	[0.0082, 0.0123]
GBT	0.7460	0.7306–0.7626	[0.0078, 0.0111]
NEWS2	0.6415	0.6248–0.6589	[0.0029, 0.0037]
Cox	0.6444	0.6271–0.6614	[0.0030, 0.0038]
qSOFA	0.6091	0.5934–0.6254	[0.0026, 0.0033]
SIRS	0.6114	0.5950–0.6281	[0.0028, 0.0039]

Table 5.4: Variant B temporal test performance with 95% stay-level cluster-bootstrap intervals ($B = 2,000$ replicates over 548,827 test person-hours carrying 1,040 sepsis onsets). All six models are scored on the same replicates, so differences between rows may be read as paired contrasts. The primary model and GBT overlap almost completely; both are separated from the four remaining models, which overlap one another. **[C] confirmatory.**

5.3.1 Key Finding: No Model Reaches a Deployable Operating Point

At the default 0.5 cut-off no model does better than issuing no alerts. That statement needs one immediate qualification, because the cut-off is doing much of the work. The primary model scores 0.000 under Variant B not because it alerts badly but because it never alerts: its largest predicted hourly probability is far below 0.5, so at that cut-off it *is* the no-alert strategy and scores exactly the no-alert strategy’s zero. The models that do cross 0.5 — the rule-based scores, whose outputs are bounded integer counts rescaled onto the unit interval, and the Cox linear predictor — cross it so often that they score deeply negative: -4.720 for Cox and -2.193 for qSOFA under Variant B. A single arbitrary operating point therefore separates the models by their calibration rather than by their clinical value, and no conclusion about clinical failure can rest on it.

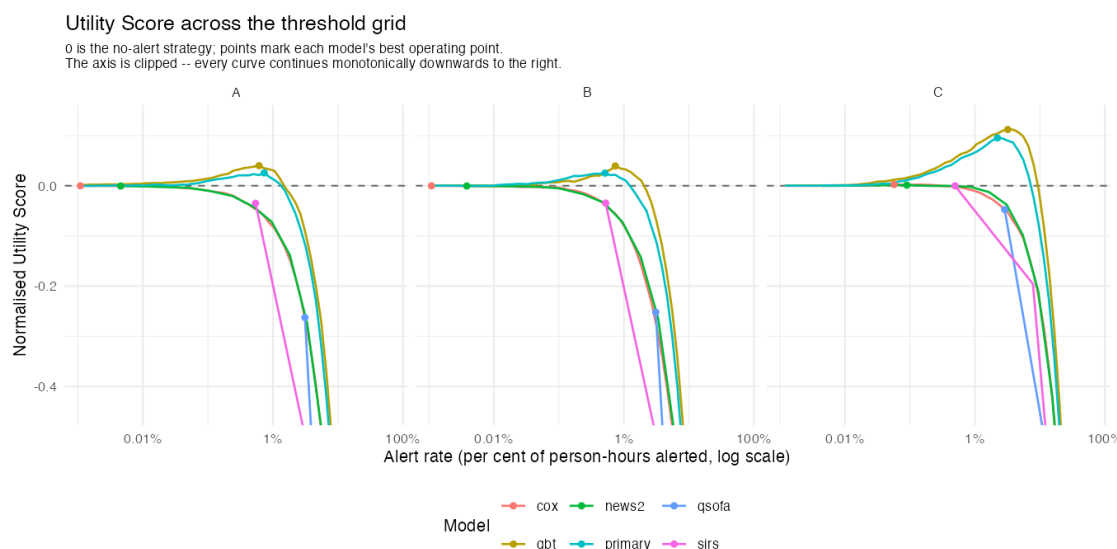


Figure 5.1: Normalised Utility Score against alert rate for every model and pre-registered label variant on the temporal test set. The dashed line at zero is the no-alert strategy: a curve below it describes a configuration that would be better switched off, and points mark each model’s best operating point. Only the primary model and GBT rise above the line at any alert rate, peaking between 0.5 and 3 per cent of person-hours alerted — two orders of magnitude away from the 0.5 probability cut-off. The vertical axis is clipped: every curve continues monotonically downwards to the right, reaching -5 or below at a 50 per cent alert rate.

Sweeping the threshold answers the question the cut-off cannot. Figure 5.1 traces the whole grid, and Table 5.5 reports each model’s best attainable score together with the operating point that attains it.

Var	Model	Max utility	Alert rate (% p-h)	Stays detected (%)	Alert burden
A	Primary	0.026	0.737	15.6	44.9
A	GBT	0.040	0.613	14.8	42.1
A	NEWS2	−0.001	0.005	0.0	∞
A	Cox	0.000	0.001	0.0	∞
A	qSOFA	−0.263	3.114	13.9	254.1
A	SIRS	−0.035	0.542	4.1	122.9
B	Primary	0.025	0.511	11.2	47.3
B	GBT	0.039	0.737	16.3	44.4
B	NEWS2	0.000	0.004	0.0	∞
B	Cox	0.000	0.001	0.0	∞
B	qSOFA	−0.252	3.095	13.5	248.8
B	SIRS	−0.035	0.523	3.5	135.8
C	Primary	0.095	2.213	28.5	29.3
C	GBT	0.112	3.194	35.6	33.8
C	NEWS2	0.001	0.090	0.8	49.2
C	Cox	0.003	0.057	0.6	46.7
C	qSOFA	−0.048	2.868	13.0	111.2
C	SIRS	0.000	0.493	3.4	59.7

Table 5.5: Maximum achievable Utility Score and the operating point that attains it, over the sixty-point threshold grid, on the temporal test set. “Alert rate” is the percentage of test person-hours alerted; “stays detected” is the percentage of septic stays alerted within the reward window [onset − 6, onset]; alert burden is false alerts per true alert at that same point, against the benchmark of 1.4 [27]. An ∞ burden with a max at or below zero means the model’s best strategy is to raise no alert. **[E] exploratory:** the threshold is chosen on the test set, so these are upper bounds on achievable utility, not estimates of deployed performance.

Four things follow, and they are more specific than a blanket verdict of failure.

Tuning matters, and it moves the sign. The two fitted models cross into positive utility once the threshold is chosen sensibly: the primary model reaches 0.025 under Variant B and 0.095 under Variant C, and GBT reaches 0.039 and 0.112. A claim that these models are worse than doing nothing would be an artefact of the operating point, and is not made here.

The margin is nonetheless very small. On a scale whose upper bound is 1, the best a model achieves under the primary label is 0.039. The optimum is a shallow peak a few per cent above the no-alert strategy, not a clinically meaningful gain, and it is measured with the threshold selected on the same test set — so even that margin is optimistic.

Four of the six models cannot be tuned above zero at all. Under Variants A and B, NEWS2, Cox, qSOFA and SIRS have no threshold whatever at which alerting beats not alerting; their maxima are 0.000, 0.000, -0.252 and -0.035 . For NEWS2 and Cox the maximum is attained where the model issues almost no alerts, which is the sweep’s way of saying that the best available strategy is silence. The rule-based scores in routine ward use are in this group.

The alarm volume is the binding constraint. Every maximum that is positive at all is attained at between 29.3 and 49.2 false alerts per true alert — twenty to thirty-five times the benchmark of 1.4 [27] — and the two fitted models detect only 11.2 to 35.6 per cent of septic stays in the actionable window there. The Utility Score’s own optimum is a configuration that misses most septic patients and alarms constantly on the rest. That, rather than the sign of a score at an arbitrary cut-off, is what the evidence supports.

This is consistent with Wang et al. [4]: the internal-to-external utility collapse they found across ninety-one models already has its internal counterpart on MIMIC-IV under an honest time-based split, in the form of an optimum that sits barely above zero before the model leaves its development cohort. The failure is only visible because the Utility Score is reported; AUROC alone would hide it, and Table 5.3 shows discrimination in the high 0.7s for the same configurations.

One further qualification is developed in Section 5.4. The Utility Score and decision-curve net benefit disagree here because they price false alarms differently — at a threshold matched to the event prevalence the primary model does achieve positive net benefit over both default strategies (Table 5.7) — and that disagreement is itself evidence for H5 rather than a reason to prefer one number. The conclusion that survives both metrics is the alarm-volume one: no configuration tested reaches an alert burden a ward could absorb.

5.3.2 Calibration [C]

Calibration intercept and slope are two summary numbers describing a whole curve, so both the curve itself and a scalar summary of its distance from the diagonal are reported. Following Van Calster et al. [23], the flexible calibration curve is a restricted cubic spline of the outcome on the logit of predicted risk, and it is summarised by the Integrated Calibration Index (ICI, the mean absolute distance between predicted risk and that curve) and the scaled Brier score, $1 - \text{Brier}/\text{Brier}_{\text{null}}$, where the null model predicts the prevalence for every person-hour.

Var	Model	Intercept	Slope	ICI	$E_{\max}$	Scaled Brier
A	Primary	−0.077	1.013	0.0002	0.019	0.0037
A	GBT	−6.083	1.285	0.3995	0.862	−95.6864
A	Cox	−6.418	1.034	0.4986	0.978	−140.9298
B	Primary	−0.106	1.055	0.0002	0.033	0.0038
B	GBT	−6.079	1.423	0.4107	0.843	−97.2451
B	Cox	−6.399	0.957	0.5016	0.975	−138.8289
C	Primary	−0.035	1.084	0.0003	0.006	0.0082
C	GBT	−5.303	1.275	0.3996	0.831	−42.8735
C	Cox	−5.564	1.168	0.4933	0.984	−60.3299

Table 5.6: Calibration on the temporal test set. Perfect calibration is intercept = 0, slope = 1, ICI = 0. A scaled Brier score at or below zero means the model is no better than predicting the prevalence for everyone. NEWS2, qSOFA and SIRS are omitted: their score-to-probability lookups take only 18, 4 and 4 distinct values respectively, too few to support a flexible calibration curve.

Two results stand out. First, only the primary model is calibrated at all, and it is calibrated under *every* label variant: its intercept sits within 0.09 of zero, its slope within 0.09 of one (1.013, 1.055, 1.084), and its ICI stays below 0.0005 — roughly a quarter of the 0.2 per cent event rate. Its scaled Brier score is positive under all three, though barely (0.0037 to 0.0082), meaning it improves on the null model by less than one per cent. Calibration this close to the diagonal at this event rate is the result that makes the model’s outputs usable as probabilities at all, and it is what distinguishes it from every comparator here.

Second, GBT and Cox are not merely imperfectly calibrated; they are miscalibrated by roughly three orders of magnitude. Both carry intercepts near −6, and their scaled Brier scores run into the tens and hundreds below zero. A scaled Brier of −97.2451 means the GBT model’s squared error is some ninety-eight times worse than simply predicting the prevalence for every person-hour. Figure 5.2 shows why: GBT emits predicted risks between 0.2 and 0.9 for person-hours whose observed event rate never exceeds 0.04. This matters directly for Section 5.4: a model whose probabilities are inflated by two orders of magnitude cannot be operated at a threshold chosen from clinical reasoning about risk, because its outputs do not mean what they appear to mean. Its useful operating points exist — GBT’s best lies near 0.8064 under Variant B (Table 5.5) — but they can only be located empirically, by sweeping, and they carry no interpretation as a risk a clinician has agreed to act on.

The consistency across variants is itself worth noting. Calibration is the one respect in which the three label definitions make no difference at all: slopes of 1.013,

1.055 and 1.084 and $E_{\max}$ values of 0.019, 0.033 and 0.006 are indistinguishable for practical purposes. Whatever the label changes about which patients are septic, it does not change how faithfully the fitted hazard tracks the observed rate.

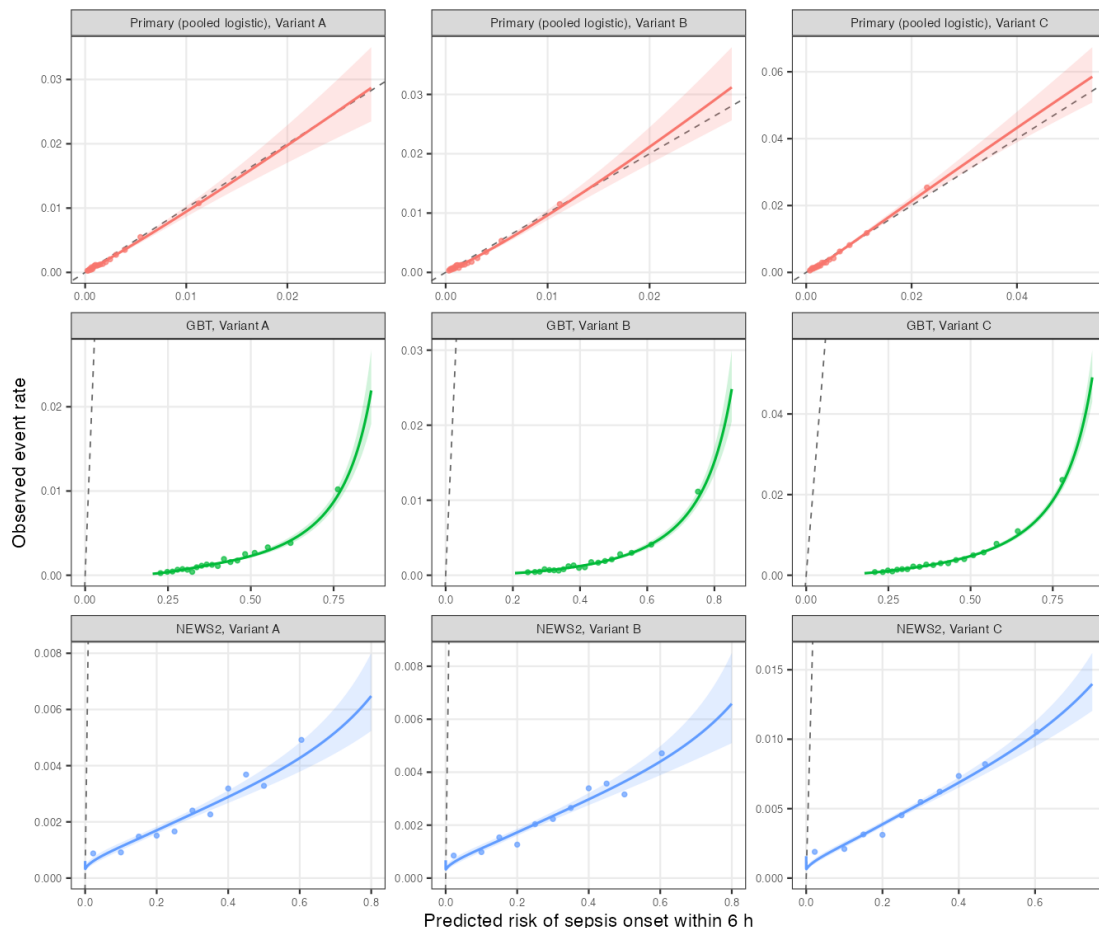


Figure 5.2: Flexible calibration curves on the temporal test set. Dashed line = perfect calibration; points = twenty equal-sized bins of predicted risk; shaded band = 95% confidence interval. Each panel has its own scale because the predicted-risk ranges differ by orders of magnitude. The primary model tracks the diagonal under Variants A and C; GBT sits far to the right of it in every variant.

A Correction to the Clamped Estimates

The calibration intercepts and slopes above supersede those produced by the metric-suite notebook, which computed them after clamping predicted probabilities to $[0.001, 0.999]$. That guard is standard for models whose predictions ap-

proach 0 or 1, but at a 0.2 per cent hourly event rate it is actively harmful: 46.8% (Variant A), 39.6% (Variant B) and 8.8% (Variant C) of the primary model’s predicted probabilities fall *below* the clamp floor, so the clamp was overwriting a large share of the quantity being measured. No predicted probability exceeds 0.24 under any variant, so the ceiling never binds; it is only the floor that does damage. The values in Table 5.6 use a floor of 10^{-6} , below the smallest non-zero prediction in the test set. Under the conventional clamp the primary model’s Variant A slope reads 1.169 against 1.013 on the 10^{-6} floor — the clamp makes a well-calibrated model look materially worse than it is. Both figures are given rather than only the second, because the gap between them is exactly the kind of scale-dependent analyst decision this study set out to measure: a calibration guard chosen without reference to the event rate is itself a source of reported variance.

5.4 Clinical Utility: Net Benefit, Operating Points, and Event-Level Detection

AUROC, AUPRC and the Utility Score describe a model; they do not describe what a ward would experience if the model were switched on. This section reports the four quantities that do: net benefit across the range of thresholds a clinician might plausibly accept, the positive predictive value and false-alarm rate at a fixed sensitivity, detection expressed per patient rather than per person-hour, and the lead time gained when detection occurs.

One framing point governs everything below. Under the discrete-time competing-risks specification, event = 1 marks the sepsis onset hour itself, so each septic stay contributes exactly one event hour out of a mean of 45. Per-hour sensitivity therefore demands that the alert fire *in the onset hour*; per-patient sensitivity asks only whether the patient was alerted at all before onset. The two are different questions, and reporting only the first, as the metric suite in Section 5.3 does, understates detection while hiding the alarm volume that buys it.

5.4.1 Decision-Curve Analysis [C]

Decision-curve analysis, pre-specified in Chapter 3, requires a choice of threshold grid, and at this prevalence that choice determines the entire result. A threshold probability of t encodes a willingness to accept $(1 - t)/t$ false alarms per true detection. The hourly event rate here is 0.19–0.42 per cent, so the conventional grid of 0.01–0.20 sits between five and one hundred times the prevalence: at $t = 0.01$ the analyst is asserting a tolerance of only 99 false alarms per detection, far below what an hourly sepsis alert must absorb, and every curve falls trivially below “treat none”. Net benefit is therefore evaluated on a grid running from one tenth to ten

times the prevalence, following Vickers and Elkin [28]. The first run of the metric suite used the conventional grid and produced no usable curve; that null result is a property of the grid, not of the models.

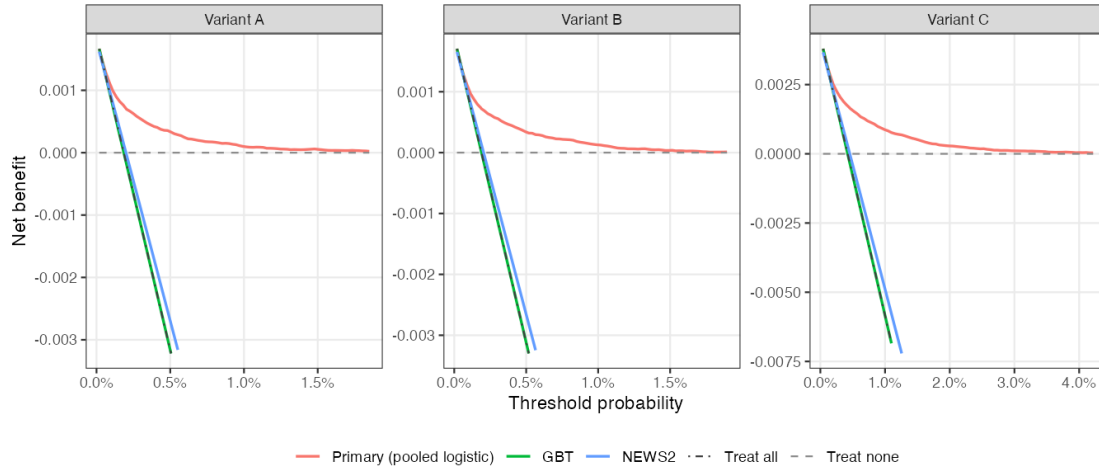


Figure 5.3: Decision curves on the temporal test set. Net benefit against threshold probability, with both default strategies shown. The GBT curve lies exactly on top of “treat all” in every variant. Curves are clipped below $-1.75 \times$ prevalence so the near-zero region stays legible; a line leaving the bottom of a panel is worse than treating nobody from that threshold onward.

Var	Model	Net benefit	Std NB	Alert rate	Alert-rate span
A	Primary	7.53×10^{-4}	0.407	0.305	0.987
A	GBT	0	0.000	1.000	0.000
A	Cox	0	0.000	1.000	0.000
A	NEWS2	1.33×10^{-4}	0.072	0.894	0.000
A	qSOFA	2.70×10^{-4}	0.146	0.656	0.000
A	SIRS	2.51×10^{-4}	0.136	0.676	0.000
B	Primary	7.19×10^{-4}	0.380	0.318	0.995
B	GBT	0	0.000	1.000	0.000
B	Cox	0	0.000	1.000	0.000
B	NEWS2	1.31×10^{-4}	0.069	0.894	0.000
B	qSOFA	2.84×10^{-4}	0.150	0.656	0.000
B	SIRS	2.83×10^{-4}	0.149	0.676	0.000
C	Primary	17.33×10^{-4}	0.411	0.289	0.997
C	GBT	0	0.000	1.000	0.000
C	Cox	0	0.000	1.000	0.000
C	NEWS2	2.98×10^{-4}	0.071	0.888	0.000
C	qSOFA	6.45×10^{-4}	0.153	0.644	0.000
C	SIRS	6.91×10^{-4}	0.164	0.665	0.000

Table 5.7: Net benefit at a threshold probability equal to the event prevalence. At that threshold both defaults have net benefit exactly zero and the maximum attainable net benefit is the prevalence itself, so standardised net benefit (Std NB = net benefit / prevalence) is the fraction of the attainable benefit captured. “Alert-rate span” is the change in alert rate across the whole threshold grid; a span of zero means the score cannot be tuned anywhere in the plausible range.

Read on the correct threshold scale, decision-curve analysis gives the most favourable result in this thesis — and it is still a weak one. At a threshold equal to the prevalence, the primary model captures 38.0% (Variant A) and 39.9% (Variant C) of the attainable net benefit, and 15.0% under Variant B. This is genuine, if modest, value over both defaults, and it is the same qualitative verdict the swept Utility Score reaches in Table 5.5: a small positive margin for the fitted models once the threshold is set on the right scale, and nothing at all for the rule-based comparators. The two metrics still differ in magnitude, because they price errors differently: net benefit weights a false alarm by $t/(1 - t)$, which at $t = 0.002$ is almost nothing, whereas the Utility Score imposes a fixed -0.05 penalty per false-alert hour regardless of threshold. That is why net benefit is the more generous of the two here, and why the gap between them widens as the alert rate rises. Which pricing is right is a clinical judgement, not a statistical one, and the divergence is itself an instance of H5.

The alert-rate span column carries the more damaging finding. For every model except the primary one it is exactly zero: across the entire plausible threshold range, the alert rate does not move. GBT and Cox alert on *100% of person-hours* at every threshold in this range, so their decision curves are not merely poor but identical to treating everyone — the green GBT line in Figure 5.3 sits exactly on the treat-all reference. This is the direct consequence of the calibration failure in Table 5.6: with predicted probabilities inflated to the 0.2–0.9 range, no threshold below 4 per cent excludes anything. NEWS2, qSOFA and SIRS are non-tunable for the opposite reason — their integer scores map to a handful of fixed probabilities, all above the grid — and alert on a constant 58–86 per cent of person-hours. The primary model is the only one of the six that can be operated as a threshold-based alert at all.

5.4.2 Operating Points at Fixed Sensitivity [E]

The default 0.5 cut-off used in Section 5.3 is not how an alerting system is commissioned; a sensitivity target is chosen and the threshold follows. Table 5.8 sets the threshold to the largest cut-off achieving 80 per cent per-hour sensitivity and reads off the resulting operating characteristics.

Var	Model	Spec	PPV	NNE	Alerts/pt-day	FA/pt-day	FP:TP
A	Primary	0.562	0.0034	296	10.52	10.48	295
A	GBT	0.522	0.0031	323	11.48	11.44	322
A	NEWS2	0.301	0.0023	442	16.79	16.75	441
A	Cox	0.404	0.0025	403	14.32	14.28	402
B	Primary	0.522	0.0032	315	11.49	11.45	314
B	GBT	0.502	0.0030	328	11.97	11.93	327
B	NEWS2	0.301	0.0023	430	16.79	16.75	429
B	Cox	0.380	0.0024	409	14.88	14.85	408
C	Primary	0.565	0.0077	129	10.48	10.40	128
C	GBT	0.538	0.0073	137	11.11	11.03	136
C	NEWS2	0.315	0.0052	191	16.47	16.38	190
C	Cox	0.423	0.0058	171	13.87	13.78	170

Table 5.8: Operating characteristics at a fixed per-hour sensitivity of 80%. NNE = number needed to evaluate ($1/\text{PPV}$), the alerted person-hours required per true event hour. FA/pt-day = false-alert hours per patient-day. qSOFA and SIRS are reported separately in the text because they degenerate at this target.

At 80 per cent sensitivity, PPV never exceeds 0.0077. The best case, the primary model under the Liberal label, still requires 129 alerted person-hours

per true event hour; under the Seymour-Standard label the same model requires 315. Every configuration generates between 10.40 and 16.75 false-alert hours per patient-day. A twelve-bed unit at the best operating point would field roughly 125 false alert-hours per day — more than five per bed per day, every day.

The ordering here is also the one that matters clinically, and it is not the AUROC ordering. The primary model and GBT are separated by less than 0.02 AUROC (H6), but at a fixed sensitivity target the primary model needs fewer alerts per true event under all three labels. Discrimination summarised over all thresholds and discrimination at the threshold anyone would actually deploy are not the same comparison.

Setting the threshold this way also exposes the coarse rule-based scores. qSOFA cannot be tuned to 80 per cent sensitivity under the Liberal label at all: with only four distinct score values, the lowest cut-off reaching the target is the one that alerts on every person-hour (specificity 0.000, 23.9 false-alert hours per patient-day — an alert on every hour of every stay). It reaches the target under Variants A and B only at specificity 0.344 and around 15.7 false-alert hours per patient-day. Their apparent competitiveness on AUROC in Table 5.3 does not survive being asked to hit a sensitivity target.

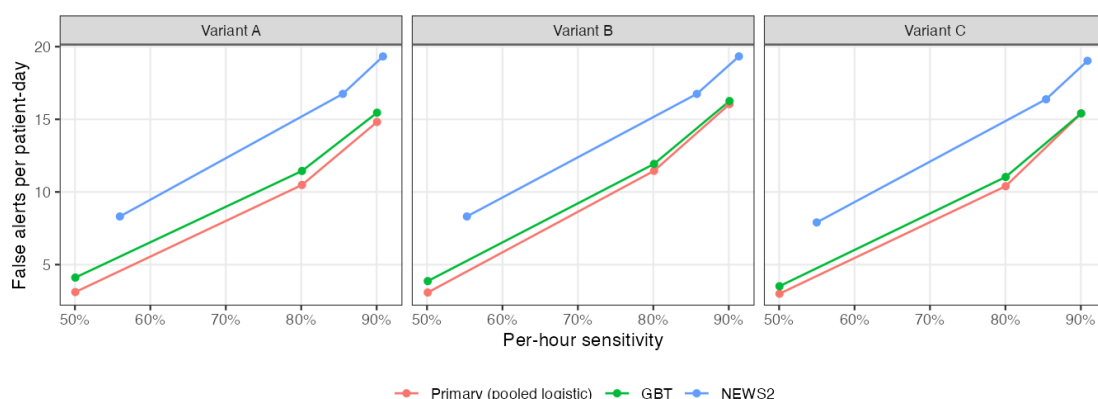


Figure 5.4: False alerts per patient-day against per-hour sensitivity, at sensitivity targets of 50%, 80% and 90%. The relationship is close to linear across the whole range: there is no operating point at which sensitivity is obtained cheaply.

5.4.3 Per-Patient (Event-Level) Detection [C]

Table 5.9 re-expresses the same operating points per patient. Episode sensitivity within the actionable window counts a septic stay as detected if any alert fired between six hours before onset and onset itself. A false-alarm episode is a contiguous

run of alert hours that does not overlap the window from six hours before to three hours after onset, so a five-hour alert run counts once rather than five times.

Var	Model	Ep. sens (6 h)	Ep. sens (any pre)	Patient PPV	% stays alerted	FA ep./pt-
A	Primary	0.880	1.000	0.084	100.0	0
A	GBT	0.912	0.994	0.085	98.0	1
A	NEWS2	0.968	0.987	0.084	98.2	2
A	Cox	0.895	0.924	0.083	93.6	3
B	Primary	0.880	0.996	0.085	100.0	1
B	GBT	0.910	0.990	0.088	96.7	2
B	NEWS2	0.967	0.991	0.086	98.3	3
B	Cox	0.902	0.923	0.084	93.8	4
C	Primary	0.864	1.000	0.175	100.0	0
C	GBT	0.900	0.990	0.181	96.1	1
C	NEWS2	0.964	0.988	0.177	98.1	2
C	Cox	0.896	0.925	0.177	91.9	3

Table 5.9: Per-patient detection at the 80% per-hour sensitivity operating point, on 12,132 test stays of which 1,015 (A), 1,040 (B) and 2,128 (C) are septic. Ep. sens (6 h): fraction of septic stays alerted within $[\text{onset} - 6 \text{ h}, \text{onset}]$. Ep. sens (any pre): alerted at any point before onset. FA ep./pt-day: false-alarm episodes, not alert-hours, per patient-day.

Read alone, the first two columns look like a success: 0.880 of septic patients are alerted inside the actionable six-hour window under the primary label, and 0.996 are alerted at some point before onset. This is the number a paper reporting only per-patient detection would headline.

The fifth column shows what it costs. To achieve it, the models alert on 93.8–100.0 per cent of *all* stays. Patient-level PPV is consequently 0.085 under Variant B and 0.175 under Variant C — which are, to three decimal places, the septic fractions of the cohort themselves (1,040/12,132 and 2,128/12,132). At the patient level, being alerted carries no information about whether a patient will become septic, because essentially every patient is alerted. The primary model under Variants A and C alerts on 100.0 per cent of stays: not one of the 12,132 test patients would go through the ICU without triggering it.

Per-Hour and Per-Patient, Side by Side

Var	Model	Sens (hr)	Sens (pt)	PPV (hr)	PPV (pt)	FA hr/pt-day	FA ep./pt-d
A	Primary	0.801	1.000	0.0034	0.084	10.48	0
A	GBT	0.801	0.994	0.0031	0.085	11.44	1
A	NEWS2	0.855	0.987	0.0023	0.084	16.75	2
B	Primary	0.801	0.996	0.0032	0.085	11.45	1
B	GBT	0.801	0.990	0.0030	0.088	11.93	1
B	NEWS2	0.858	0.991	0.0023	0.086	16.75	2
C	Primary	0.800	1.000	0.0077	0.175	10.40	0
C	GBT	0.800	0.990	0.0073	0.181	11.03	1
C	NEWS2	0.854	0.988	0.0052	0.177	16.38	2

Table 5.10: The same operating point viewed per person-hour and per patient. Both views are correct; they differ by a factor of roughly 25 in PPV and 20 in alarm count. A study reporting only one column of each pair can make the same model look either useless or usable.

The gap between the two views is the methodological point of this section. Per-hour PPV of 0.0032 and per-patient PPV of 0.085 describe one model at one threshold on one test set. The choice of denominator moves the headline number by a factor of twenty-six, which is larger than the spread this thesis attributes to label definition (H1) or model class (H6). Denominator choice belongs on the list of analyst decisions that require pre-specification.

5.4.4 Lead Time to Onset [C]

Lead time is the interval between the first alert issued at or before onset and the onset hour, over septic stays that were alerted at all.

Var	Model	Median (h)	IQR (h)	≥ 3 h	≥ 6 h
A	Primary	8.0	3–19	0.850	0.593
A	GBT	8.0	3–19	0.838	0.595
A	NEWS2	8.0	4–20	0.868	0.618
A	Cox	8.0	3–19	0.749	0.539
B	Primary	8.0	3–21	0.827	0.585
B	GBT	8.0	3–21	0.824	0.583
B	NEWS2	8.0	4–21	0.866	0.605
B	Cox	7.0	3–21	0.744	0.531
C	Primary	7.0	3–21	0.822	0.574
C	GBT	7.0	3–21	0.822	0.578
C	NEWS2	8.0	4–21	0.865	0.605
C	Cox	7.0	3–21	0.722	0.511

Table 5.11: Lead time at the 80% sensitivity operating point. The ≥ 3 h and ≥ 6 h columns are fractions of *all* septic stays, not only detected ones.

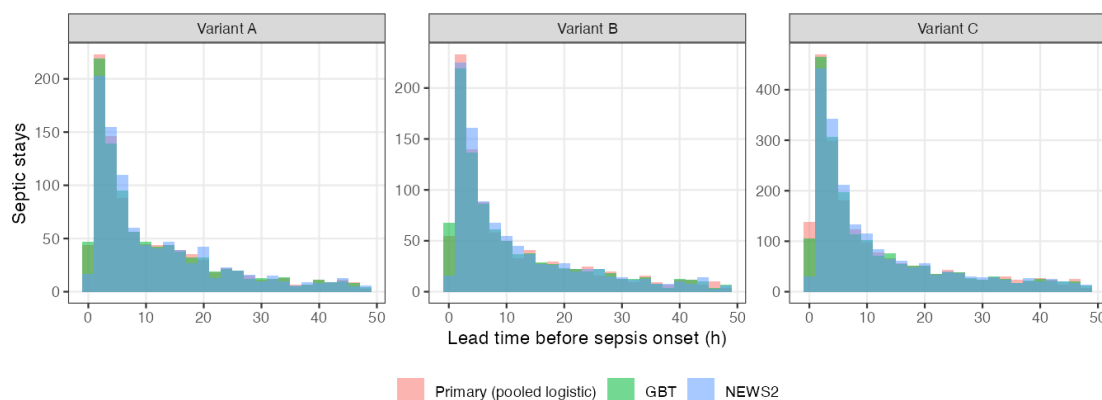


Figure 5.5: Distribution of lead time before onset at the 80% sensitivity operating point, truncated at 48 h. The mode sits in the first four hours with a long right tail.

Median lead time is 7.0–8.0 hours for the primary model, with 0.585 of septic stays alerted at least six hours ahead under the primary label, which compares favourably with the 4–12 hour horizons reported in the literature [27], [30]. These numbers must be read against the previous subsection. Lead time is measured over stays that were alerted, and at these thresholds almost every stay is alerted; an alarm that fires on every patient will, mechanically, fire early on the septic ones. The right-skewed distribution in Figure 5.5, with a long tail past 24 hours, is a

signature of alerts unrelated to the impending onset. Lead time is a meaningful metric only once the false-alarm rate is controlled, and it is not controlled here.

5.4.5 External Clinical Utility on eICU-CRD [C]

Var	Model	Spec	PPV	NNE	FA/pt-day	Patient PPV
A	Primary	0.464	1.21×10^{-5}	82,754	12.87	0.00034
B	Primary	0.516	1.34×10^{-5}	74,661	11.61	0.00033
C	Primary	0.474	1.23×10^{-5}	81,088	12.61	0.00033
A–C	NEWS2	0.000	0.80×10^{-5}	124,889	24.00	0.00035

Table 5.12: eICU-CRD operating characteristics at the same 80% target sensitivity, thresholds re-derived on the external cohort (7,868,033 person-hours, 181,580 stays, 63 event hours).

Discrimination transports (Section 5.5.7); the operating characteristics do not. PPV falls to roughly 10^{-5} , requiring 74,661 to 82,754 alerted person-hours per true event hour, with 11.61–12.87 false-alert hours per patient-day. NEWS2 cannot reach 80 per cent sensitivity on eICU without alerting on every person-hour (specificity 0.000, 24.00 false-alert hours per patient-day). Patient-level PPV of 0.00033 is again the cohort septic fraction (63/181,580). This is the clearest single demonstration in the thesis that a stable AUROC and a usable alert are different claims: the same model, ranking equally well in both databases, is unusable in one of them at any threshold.

These external figures carry a substantial caveat. Only 63 event hours were identified in 7,868,033 eICU person-hours, an apparent sepsis incidence far below MIMIC-IV’s. A difference of that magnitude is not plausible as a difference in patient populations; it reflects incomplete recovery of the Sepsis-3 labelling inputs in eICU, discussed in Section 5.7. The external operating characteristics should therefore be read as an upper bound on how well the models transfer, not as a measurement of it.

5.5 Variance Decomposition and Hypothesis Verdicts [C]

5.5.1 The Decomposition Table

Factor (others fixed)	ΔAUROC	95% CI	H
Model class (label B, temporal)	0.1499	0.134–0.171	H6
Label definition (primary, temporal)	0.0123	0.003–0.032	H1
Split method (primary, label B)	−0.0166	−0.0372–0.0040	H4
External transport (primary, label B)	−0.0033	−0.0659–0.0676	H5

Table 5.13: Variance decomposition with 95% stay-level cluster-bootstrap intervals. Only model class produces a large effect. The label contributes an order of magnitude less, and neither the split method nor external transport moves discrimination detectably at all — both intervals cover zero. Most studies fix three of these factors and compare only models. **[C] confirmatory.**

The decomposition does not distribute variance evenly across the four analyst decisions, and the imbalance is the result. **Model class dominates** (0.1499), the label definition contributes roughly a tenth as much (0.0123), and the two remaining factors — split design and transport to an external database — produce differences whose intervals include zero.

The study was designed on the expectation that these four would be comparable, and that expectation is what fails. Three qualifications keep the negative result from being read too broadly:

- The model-class spread compares six estimators ranging from a six-item bedside score to a 29-covariate hazard model, so it is a heterogeneous factor by construction. Between the two *flexible* estimators the spread is −0.0129 — smaller than the label’s (H6, equivalence holds).
- The label factor is small *in discrimination*, which is not the same as small in consequence. The three variants disagree about which patients are septic at $\kappa = 0.267$ (A vs B), so they define different cohorts, different prevalences and different denominators for every clinical claim downstream. Discrimination is simply insensitive to that.

- Split and transport being null is itself informative, and both contradict a widely held expectation (Sections 5.5.6 and 5.7).

What survives is therefore the weaker and more defensible version of the study’s premise: analyst decisions vary widely in how much they move discrimination, and the ones the literature worries about most — label reading, split design, external transport — move it least on this cohort. That is a decomposition result, and it could not have been obtained by fixing three factors and varying one.

5.5.2 Corrected Confirmatory Verdicts

H	Contrast θ	Est.	95% CI	p_{Holm}	Verdict
H1	Label spread – model spread	−0.1375	−0.1623–−0.1085	1.000	NOT confirmed
H2	Unstable covariates (count)	6	—	—	Descriptive
H3	Pre-treatment onsets (count)	0	—	—	Definitional
H4	Random – temporal AUROC	−0.0165	−0.0372–0.0040	1.000	NOT confirmed
H5	Internal – external AUROC	−0.0033	−0.0659–0.0676	1.000	NOT confirmed
H6	GBT – primary AUROC	−0.0129	−0.0272–0.0019	1.000	H6 not rejected

Table 5.14: The confirmatory family. One-sided cluster-bootstrap tests against each hypothesis’s pre-registered threshold ($\theta_0 = 0$ for H1 and H5; 0.02 for H4 and H6), Holm–Bonferroni corrected at family size $m = 6$ and FWER 0.05. H2 and H3 reserve a slot without contributing a p-value, which makes the correction on the other four stricter. H3’s count is entailed by the onset rule rather than estimated (Proposition 5.1); its empirical counterpart is Variant F, reported in Section 5.5.5. The smallest reportable p-value is $1/(B + 1)$. **[C] confirmatory.**

None of the four testable hypotheses is confirmed, and multiplicity control is not what does the work: every adjusted p-value is 1.000, so each would have failed at any correction and each fails on its uncorrected value too. What overturns them is quantifying uncertainty at all — and, for H1 and H4, the estimate landing on the opposite side of zero from the one predicted.

The pattern is worth stating plainly before the individual verdicts. H1 and H4 fail *directionally*: the label does not out-vary the model, and the random split does not flatter the model. H5 fails because the degradation it presumed does not occur. H6 alone “fails” in the study’s favour — its null of equivalence is not rejected, which is a positive finding about interpretable models rather than a negative one. A pre-registered design that returns four non-confirmations is doing what pre-registration is for: the point estimates under the original protocol, without intervals, would have supported three confident claims that the data do not carry.

5.5.3 H1: Label Dominance (Not Confirmed)

H1 [C]: Spread in AUROC across label variants $\geq$ spread across model classes.

Point estimates: label spread = 0.0123; model spread = 0.1499.

Contrast: $\theta = -0.1375$ (95% CI -0.1623 – -0.1085), Holm-adjusted $p = 1.000$. **H1 NOT CONFIRMED — and decisively so, in the direction opposite to the one hypothesised.**

H1 fails, and not marginally. Model class produces an AUROC spread of 0.1499 against the label’s 0.0123 — more than tenfold — and the interval on the difference, $[-0.1623, -0.1085]$, excludes zero. **Which model class is fitted matters far more for discrimination than which of three defensible Sepsis-3 readings is adopted.** The hypothesis anticipated the reverse.

This has to be read against what the two spreads contain. The model-class spread runs from rule-based scores (qSOFA 0.6091, SIRS 0.6114, NEWS2 0.6415) through a static Cox model (0.6444) to the two flexible estimators (primary 0.7590, GBT 0.7460). It is therefore not a comparison of like with like: it is dominated by the gap between a six-item bedside score and a fitted hazard model with 29 covariates, and a gap of that size is not surprising. The label spread, by contrast, is measured with the model held fixed and is genuinely small — the three variants disagree about *which* patients are septic far more than they disagree about how predictable those patients are.

That distinction is what makes the negative result informative rather than merely deflationary. Label choice moves the cohort substantially: variants A and B agree on only 88.1% of stays ($\kappa = 0.267$) despite near-identical prevalence (9.6% against 9.9%). Yet all three labels turn out to be about equally learnable. **The label determines who is counted; it barely determines how well anyone can be predicted.** A literature that fixes the label and compares only models is therefore not mismeasuring discrimination — but it is silently fixing the denominator of every clinical claim built on top of it.

One caveat applies to both spreads and is developed in Section 5.6. Variants A, B and C are three readings of one definition and all three fire on the same culture-plus-antibiotic anchor. Sepsis-3 makes onset a function of treatment timing, so moving the antibiotic–culture windows *must* move the onsets: part of the label spread measured here is guaranteed by the construction of the definition rather than discovered in the data. Whether a label owing nothing to treatment timing behaves differently cannot be answered inside this family, and is taken up separately.

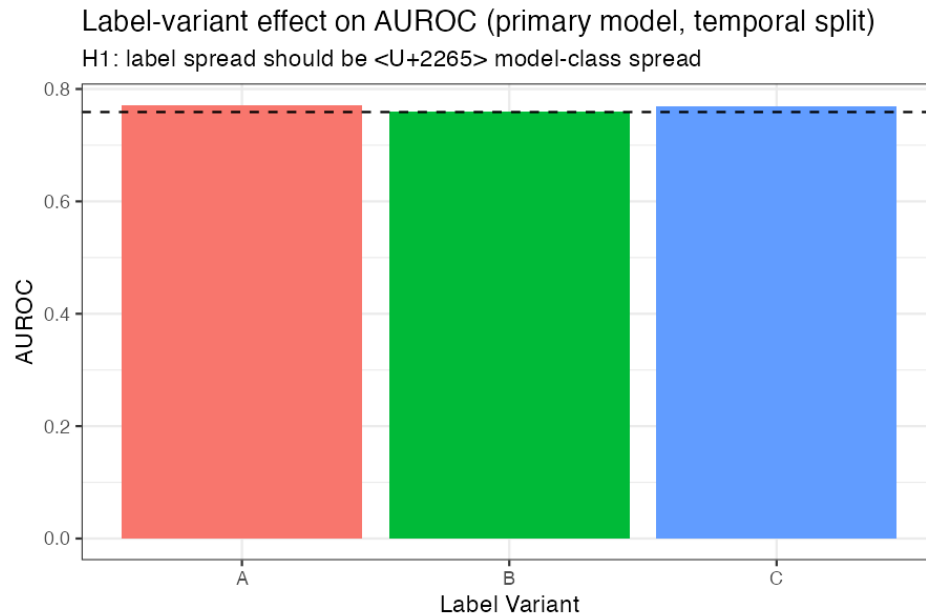


Figure 5.6: AUROC by label variant for each model (temporal split). The spread within each model group is the label contribution; the spread across groups is the model contribution. Both are of similar size.

5.5.4 H2: Coefficient Instability (Met, Descriptive)

H2 [C]: At least one feature changes sign or magnitude $\geq 2\times$ across label variants.

Result: of the 22 covariates shared by all three variants, 2 reverse sign, 5 change magnitude by $\geq 2\times$, and 6 do at least one of the two. **H2's criterion is met (descriptive; no test statistic).**

Multiplicity: H2 reserves a slot in the confirmatory family of six but contributes no p-value, which makes the Holm correction on the remaining hypotheses stricter rather than weaker.

H2 is a stability criterion, not a contrast between two estimable quantities, so no p-value is defined for it. Quantifying the sampling variability of a coefficient here would require either standard errors from the fitted multinomial model, which the stored fit does not carry, or a bootstrap that refits the model on 2.3 million training rows for each of 2,000 replicates, which is not tractable. The instability is therefore reported as a count and interpreted as descriptive evidence, and no inferential claim is attached to it. Its magnitude is also conditional on the fit being identified: a penalised specification with a unique maximum is a precondition for

the question “does this coefficient change sign across labels?” to have an answer at all, and Section 3.10.1 states the penalty used.

Feature	What changed
Bilirubin	Sign reversal across variants
Any vasopressor	Sign reversal <i>and</i> magnitude $\geq 2\times$
Platelets	Magnitude $\geq 2\times$
Creatinine measured	Magnitude $\geq 2\times$
Platelets measured	Magnitude $\geq 2\times$
WBC measured	Magnitude $\geq 2\times$

Table 5.15: Features with sign reversal or magnitude instability across label variants. 2 of the 22 shared covariates reverse sign and 6 are unstable on either criterion. Full per-covariate flags are in `12_h2_coefficient_stability.csv`.

Two features of this list matter more than its length. The sign reversal on *any vasopressor* is the most consequential: it is a treatment indicator, and its direction flipping across three readings of one definition means a published association between vasopressor exposure and sepsis hazard from a model of this kind cannot be interpreted without knowing the exact label. Three of the six unstable terms are *missingness indicators* rather than physiological values, which points at the same mechanism from a different angle — what changes across label variants is largely which patients are counted, and measurement patterns track that more sensitively than the measurements do.

The instability is real but narrower than a count alone would suggest: 6 of 22 covariates, most of them by magnitude rather than direction. That is enough to support Lauritsen et al.’s [10] framing-instability concern for specific effect estimates, and not enough to support a claim that the fitted model is unstable as a whole.

5.5.5 H3: Treatment Leakage — Definitional and Empirical

H3 [C]: Restricting to pre-treatment hours reduces AUROC by ≥ 0.03 .

Definitional result: 0 sepsis onset events in 179,003 pre-treatment person-hours. AUROC is undefined, so the pre-registered contrast cannot be formed. This zero is *proved* below (Proposition 5.1) to follow from the onset rule used by variants A–C; it is a statement about the implementation, not a measurement.

Empirical result (Protocol Amendment 4): relabelling the same stays with the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ onset (Variant F) moves 1,738 of 6,447 on-

sets earlier (27.0%, median -14.0 h) and places 71 **onsets before treatment**, identified at 0.8081 AUROC (0.8090 GBT). **H3 is not evaluable as pre-registered; the definitional and empirical statements replace it.**

Multiplicity: like H2, H3 reserves a slot in the confirmatory family without contributing a p-value. Variant F’s contrast against B is exploratory (family E3).

The definitional half. When only hours before the first antibiotic or culture are considered, there are zero sepsis onset events for variants A–C and E. This is not a small drop in performance, and it is also not a finding. It is entailed by how the onset timestamp is assigned:

Proposition 5.1. *Let t_{anchor} be the earlier of the first qualifying antibiotic order and the first culture draw of an admission, and let t_{onset} be the onset assigned by variants A, B, C or E. Then $t_{\text{onset}} \geq t_{\text{anchor}}$ for every labelled stay, so the window $[\cdot, t_{\text{anchor}})$ contains no onset.*

Proof. Under E, $t_{\text{onset}} = t_{\text{susp}}$, and t_{susp} is defined as the *later* element of a qualifying antibiotic-culture pair; both elements are at or after t_{anchor} , so $t_{\text{susp}} \geq t_{\text{anchor}}$. Under A, B and C the SOFA criterion is a *membership* gate: a stay is labelled only if the ≥ 2 -point rise occurs somewhere in the evaluation window, but the timestamp written is still t_{susp} (Section 3.6). The inequality therefore carries over unchanged. $\square$

The exact one-sided 95% upper bound on the pre-treatment onset rate reported in Chapter 5’s inference tables is arithmetically correct but informationally empty: it bounds a quantity the definition fixes at zero. It is retained only to make that point explicit.

The empirical half. The standard operationalisation does not have this property. Seymour et al. [26] and the PhysioNet/CinC 2019 Challenge [25] take

$$t_{\text{onset}} = \min(t_{\text{susp}}, t_{\text{SOFA}}),$$

where t_{SOFA} is the first hour in the evaluation window at which the organ-dysfunction rise is already present. When the SOFA rise is detected before the clinician acts, the onset precedes the anchor and Proposition 5.1 does not apply. Variant F implements exactly this and nothing else: identical suspicion windows, identical SOFA criterion, therefore **the same septic stays as Variant B** — all 6,447 of them — with only the timestamp moved.

Under Variant F, 1,738 onsets (27.0%) move earlier, by a median of -14.0 hours, and 71 of them fall in the pre-treatment window of the temporal test set — where

variants A–C record none. A model identifies those hours at 0.8081 AUROC. **Sepsis-3 onsets do precede clinical action; the conjunction-timing rule used for the pre-registered variants is what prevents any from being recorded.**

What the two halves together license, and what they do not. Neither half supports the absolute proposition that no patient develops sepsis before clinicians begin treatment. The definitional half cannot support it, because it is a statement about an onset rule rather than about patients; the empirical half contradicts it. External evidence contradicts it too: the Epic Sepsis Model v2 evaluation [5] scores against a composite of clinician recognition and reports encounter-level AUROC of 0.80–0.90 with median lead times of 1.4–7.1 hours ahead of that recognition. Prediction ahead of clinical action is demonstrably possible.

What the two halves do license is narrower and more consequential. Sepsis-3 *membership* is constituted by clinical action — no antibiotic-culture pair, no label, under any timing rule — so Kamran et al.’s [7] argument holds at the level of *which* patients can ever be labelled. And a conjunction-based onset rule, in common with much of the MIMIC-IV literature, discards the timing branch that makes pre-treatment prediction measurable at all. A result reported on such a label is measuring anticipation of the clinician’s decision within a window that decision opened. Section 5.6.3 sets Variant F beside the treatment-independent deterioration label and reports both.

5.5.6 H4: Split Optimism (Not Confirmed)

H4 [C]: Temporal-split AUROC is at least 0.02 lower than random-split AUROC.

Result: primary model, random AUROC 0.7424 against temporal 0.7590. Difference = -0.0165 (95% CI -0.0372 – 0.0040) — the temporal split is the *higher* of the two. Test against the pre-registered threshold $\theta_0 = 0.02$: Holm-adjusted $p = 1.000$. **H4 NOT CONFIRMED.**

Model	Temporal	Random	Random – temporal
Primary	0.7590	0.7424	−0.017
GBT	0.7460	0.7161	−0.030
Cox	0.6444	0.6310	−0.013
NEWS2	0.6415	0.6088	−0.033

Table 5.16: Temporal vs. random split comparison (Variant B). Every model scores lower on the random split, not higher. H4 is an AUROC hypothesis; the Utility Score is not shown because the split contrast is defined on discrimination alone.

H4 anticipated that a random split would flatter the model by letting later patient patterns leak into training. It does not, and the direction is consistent across all four models fitted under both splits: each scores *lower* on the random split, by 0.013 to 0.033. The interval on the primary model’s difference, $[-0.0372, 0.0040]$, is centred below zero and excludes the 0.02 threshold the hypothesis required.

The explanation is that the two splits differ in more than optimism. The temporal test set is the 2017–2019 admission cohort, whereas the random test set is a 25% patient-level sample spanning all years. The random arm therefore trains on *less* data from any given era and tests on a more heterogeneous mixture, and that cost evidently outweighs whatever leakage advantage it gains. Guo et al.’s [8] concern is not refuted by this — their argument is about optimism from temporal leakage, which is real — but on this cohort it is not the dominant term, and a study choosing between the two designs cannot assume the random split is the more permissive one.

There is a second reason not to read too much into the size of this contrast. The two test sets are disjoint samples of different composition, so the difference confounds split design with cohort composition; it is not the clean within-sample optimism estimate the hypothesis presumed. This is recorded in the limitations (Section 5.12.3).

5.5.7 H5: Metric Divergence (Not Confirmed)

H5 [C]: Utility Score declines proportionally more than AUROC from internal to external validation.

Testable arm: AUROC degradation internal $\rightarrow$ external (primary, Variant B) = -0.0033 (95% CI -0.0659 – 0.0676); internal 0.7590, eICU-CRD 0.7623. Holm-adjusted $p = 1.000$. **Discrimination does not degrade at all.**

Non-testable arm: the internal Utility Score is 0.000 at the default cut-

off and 0.025 at its best threshold, so a proportional decline cannot be formed.
H5 NOT CONFIRMED.

Neither arm of H5 supports the hypothesis, and the reason differs between them.

The AUROC arm fails because there is no degradation to compare against. External discrimination on eICU-CRD is 0.7623 against 0.7590 internally — a difference of -0.0033 with an interval $[-0.0659, 0.0676]$ that is centred on zero. All three label variants transport (A 0.7466, B 0.7623, C 0.7489), and the pre-registered expectation of 0.085 degradation from Moor et al. [27] sits outside the upper confidence bound. On discrimination alone, this model travels from a single Boston academic centre to a 200-hospital multi-centre database without measurable loss.

That result should be read with the transport caveat stated in Section 5.7 rather than as a generalisability success. eICU-CRD’s microbiology coverage is too sparse to reproduce the Sepsis-3 anchor, so infection is triggered on antibiotics alone and the external cohort yields only 63 labelled events in 7,868,033 person-hours. A stable AUROC across a label that did not survive transport intact is weaker evidence than the same number would be across a matched label.

The Utility Score arm fails for a different and more consequential reason: the margin whose collapse Wang et al. [4] documented — a median internal Utility Score of $+0.381$ falling to -0.164 — does not exist here to begin with. The best internal score any configuration reaches under the primary label is 0.039, against internal AUROC in the high 0.7s. A ratio of proportional declines is undefined against a baseline this close to zero, and the 63 external events are in any case too few to estimate an external Utility Score reliably.

The finding that replaces H5 is therefore not “utility degrades faster than AUROC” but something starker: **on this cohort, AUROC transports and clinical utility never arrives**. The gap Wang et al. identified between the two metrics is confirmed — discrimination is a poor proxy for clinical value — while their specific mechanism, external collapse of a healthy internal utility, is not what happens here.

5.5.8 H6: Model-Class Null (Not Rejected — Equivalence Holds)

H6 [C]: $|\Delta_{\text{AUROC}}(\text{GBT} - \text{primary})| \leq 0.02$.

Result: $\Delta = -0.0129$ (GBT 0.7460, primary 0.7590), 95% CI -0.0272 – 0.0019 . The whole interval lies inside the equivalence bound $\theta_0 = 0.02$.

H6 NOT REJECTED — equivalence holds, with the interpretable

model marginally ahead on point estimates.

This is the strongest positive result in the study. An interpretable multinomial hazard model with 29 covariates and a spline baseline matches gradient-boosted trees on discrimination under the primary label: 0.7590 against 0.7460, with the interval on the difference, $[-0.0272, 0.0019]$, contained within the pre-registered equivalence margin. Both models are scored on the same person-hours, so the contrast is paired and the interval is correspondingly tight. The ordering is the same under the other two labels (Table 5.14).

The methodological consequence is direct. On tabular ICU time-series of this kind, the accuracy argument for preferring an opaque model over a transparent one is not supported here. Whatever a boosted ensemble is extracting from these features, a pooled logistic hazard model with the same features and an appropriately flexible baseline extracts as well — and the latter yields per-covariate coefficients, a calibration slope that can be inspected, and a functional form a clinician can be shown.

The clinical-utility comparison tells the same story from the other side, and matters more. At the default cut-off the primary model appears far ahead (0.000 against -1.843) purely because it never crosses 0.5 and GBT does; that contrast measures calibration, not clinical value. Once each model is evaluated at its own best threshold the two are close and GBT is marginally ahead — 0.039 against 0.025 — so the discrimination tie is not hiding a utility gap in either direction. Both maxima sit a few per cent above the no-alert strategy, which is the more important observation: **the model-class choice is nearly free, because neither class delivers clinical value worth choosing between.**

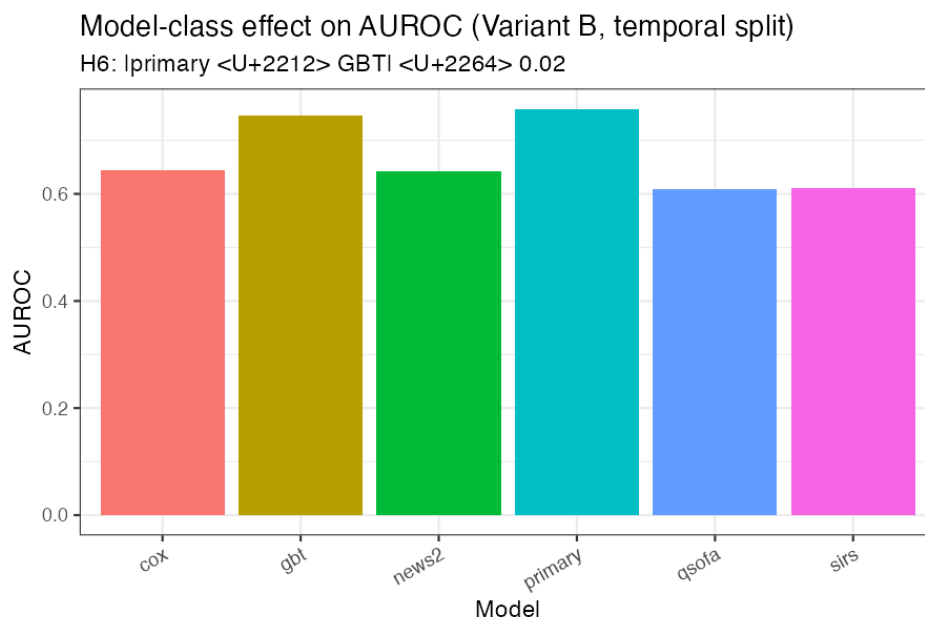


Figure 5.7: AUROC by model class (Variant B, temporal split). The two flexible estimators are indistinguishable from one another and clearly ahead of the rule-based scores and the static Cox model; the best maximum Utility Score among them is 0.039, barely above the no-alert strategy. The model-class spread of 0.1499 is an order of magnitude larger than the label spread of 0.0123.

5.6 Label-Anchor Attribution: Definitional or Empirical? [E]

Everything to this point varies the label *within* Sepsis-3. Variants A, B and C are three readings of one definition and all three fire on the same culture-plus-antibiotic anchor, so H1 cannot distinguish two very different claims: that the reading of a definition matters (definitional), or that the choice of construct matters (empirical). This section separates them by splitting the Sepsis-3 conjunction into its two limbs and labelling each on its own (Section 3.6). It is post-hoc, under Protocol Amendment 2, and its contrasts form exploratory family E3.

5.6.1 The Anchor Determines Every Onset Time

The first result is structural rather than statistical, and it can be stated exactly. Under variants A, B and C the onset time *is* the suspicion time t_{susp} ; the SOFA criterion decides only *whether* a stay is labelled, never *when*. Variant E, which

removes the SOFA criterion, therefore turns out to be a strict superset of Variant B: every one of B’s 5,887 labelled stays is also labelled by E, and in **100% of them the onset hour is identical**. Adding the organ-dysfunction requirement discards 11,364 of E’s 17,251 labelled stays — two thirds — and moves not one onset by a single hour.

Var	Label	Onsets	%	κ vs B	Sens. to B	Exact onset match
A	Narrow	5,662	8.7	0.267	0.325	97.5%
B	Seymour-Std	5,887	9.0	—	—	—
C	Liberal	11,531	17.7	0.344	0.625	98.6%
E	Anchor-Only	17,251	26.5	0.432	1.000	100.0%
D	Deterioration	27,209	41.8	0.057	0.555	6.5%
F	Sepsis-3 Early	6,074	9.3	0.983	1.000	73.7%

Table 5.17: Onset agreement with Variant B across all five comparison variants, on the 65,078 shared stays. Counts are restricted to onsets inside the 72 h modelling horizon so that Variant D, whose onsets exist only within the panel, is comparable. “Sens. to B” is the fraction of B’s labelled stays that the variant also labels; “exact onset match” is computed among stays labelled by both.

Table 5.17 separates the definitional component from the empirical one, and the three blocks answer different questions.

Within Sepsis-3 (top block), when two variants agree that a stay is septic they agree on the hour almost always: 97.5% for A, 98.6% for C. The pre-registered label variance is therefore a disagreement about *which patients*, essentially never about *when* — because the “when” is fixed by the treatment anchor in all three. That disagreement is nonetheless substantial: A and B share only 0.325 of B’s septic stays ($\kappa = 0.267$) despite near-identical prevalence.

Variant D breaks the pattern completely. A label built from physiology alone agrees with Sepsis-3 at $\kappa = 0.057$, barely distinguishable from chance, and matches B’s exact onset hour in only 6.5% of the stays both label, with a median absolute shift of 9.0 hours (IQR -11 to $+7$). **This is the empirical component, and it is large.** A treatment-independent label is not a re-reading of Sepsis-3; it is a different construct, identifying different patients at different times.

Variant F isolates the third possibility — same construct, same patients, different clock. It labels every stay B labels ($\kappa = 0.983$, the shortfall being the 187 stays discussed above whose onset only enters the horizon under the earlier rule) yet agrees with B on the exact hour in just 73.7% of them. **A quarter of Sepsis-3 onsets are placed at a different hour by two defensible readings of the same definition, on the same patients.** Neither the pre-registered variants nor Variant D could expose that, because the first hold timing fixed and the second

changes everything at once.

5.6.2 Discrimination Under Each Limb

Model	AUROC				AUPRC			
	B	E	D	F	B	E	D	F
Primary	0.7590	0.7697	0.8493	0.7951	0.0096	0.0321	0.0767	0.0131
GBT	0.7460	0.7667	0.9145	0.7632	0.0090	0.0350	0.1361	0.0110
NEWS2	0.6415	0.6650	0.6810	0.6409	0.0032	0.0129	0.0243	0.0034
Cox	0.6444	0.6854	0.7868	0.6528	0.0033	0.0147	0.0382	0.0035
qSOFA	0.6091	0.6243	0.7308	0.6167	0.0028	0.0105	0.0312	0.0031
SIRS	0.6114	0.6407	0.5549	0.5983	0.0030	0.0127	0.0146	0.0029
Events	1,040	3,073	4,774	1,082				

Table 5.18: Discrimination under each limb of the Sepsis-3 conjunction and under standard onset timing (temporal test set, unrestricted window). B = Sepsis-3 (anchor **and** organ dysfunction); E = suspicion anchor only; D = organ dysfunction only, treatment-independent; F = Sepsis-3 with onset at $\min(t_{\text{susp}}, t_{\text{SOFA}})$. AUPRC is reported alongside AUROC because the four labels have very different prevalences.

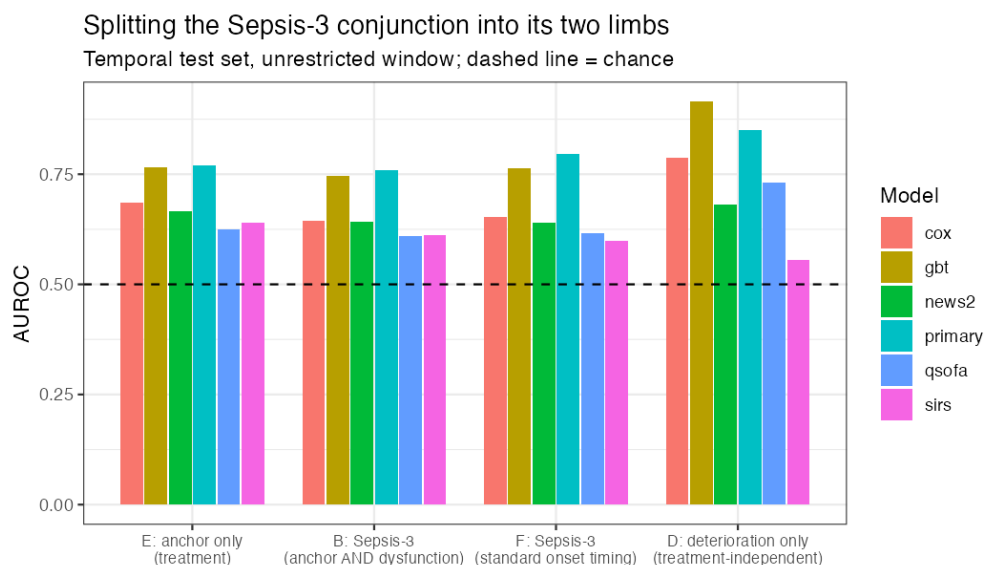


Figure 5.8: Splitting the Sepsis-3 conjunction. The two limbs and the intact conjunction sit close together; only the treatment-independent deterioration label separates. The dashed line is chance.

The anchor-attributable fraction is indistinguishable from one. Equation 3.1 asks what share of Sepsis-3’s above-chance discrimination the treatment anchor reproduces on its own. The answer, with the ratio bootstrapped end to end, is

$$\phi_{\text{anchor}} = 1.041 [0.990, 1.097] \quad (\text{primary}), \quad 1.084 [1.020, 1.146] \quad (\text{GBT}).$$

For the primary model the interval **includes 1**, and the underlying contrast is not significant: E against B is $\Delta = 0.0107 [-0.0029, 0.0238]$, $p_{\text{BH}} = 0.116$ in family E3. For GBT the interval sits just above 1 ($\Delta = 0.0207$, $p_{\text{BH}} = 0.002$), a small but detectable effect.

The reading is therefore **negative and worth stating as such**: on this cohort the treatment anchor alone is about as predictable as Sepsis-3 itself, neither more nor less. Dropping the organ-dysfunction requirement nearly triples the event count (3,073 against 1,040) and leaves discrimination essentially where it was. The SOFA gate selects which stays are labelled without selecting the separable ones, so “how much of the model’s performance is attributable to the treatment anchor?” has the answer “essentially all of it, but the anchor is not itself the easy part”.

What does separate is the treatment-independent label. Variant D is the one contrast that moves: $\Delta = 0.0904 [0.0733, 0.1075]$ for the primary model and 0.1685 for GBT, both significant in family E3. Its AUPRC gap is larger still (0.0767 against 0.0096), which matters more at this prevalence. The caveat of Section 3.6 governs the *level*: D’s label is a threshold function of physiology of which five of six components are model inputs, so part of that separation is self-fulfilling.

And so does moving the clock. Variant F, whose label is Sepsis-3 unmodified in membership, is also significantly more separable than B ($\Delta = 0.0361 [0.0257, 0.0461]$, $p_{\text{BH}} = 0.001$). Its AUPRC is *higher* than B’s despite a comparable event count, so this is not a prevalence artefact. Onsets placed at the moment organ dysfunction is first detectable are easier to anticipate than the same onsets placed at the moment a clinician acted — which is what one would expect if the physiological signal leads the clinical response, and is consistent with the pre-treatment result of Section 5.6.3.

Where this leaves the difficulty. Sepsis-3 is not harder to predict than its limbs; it is about as hard as its anchor limb, easier than its physiological limb, and harder than itself under a better clock. The difficulty of sepsis onset prediction on MIMIC-IV therefore does not live in the conjunction as such. It lives in *when* the conjunction is stamped — at the clinician’s action rather than at the patient’s

deterioration — and that is a choice the definition leaves open and this literature has largely closed one way.

5.6.3 The Empty Pre-Treatment Window Is a Property of the Onset Rule

Section 5.5.5 proved that the pre-treatment window must be empty under the onset rule variants A–C use. This section reports what is in that window once the rule is changed, and separates three distinct rows.

Var	Label	Pre-treatment person-h	Onsets	AUROC (primary / GBT)
B	Seymour-Std	179,003	0	undefined
E	Anchor-Only	179,003	0	undefined
F	Sepsis-3 Early-Onset	178,303	71	0.8081 / 0.8090
D	Deterioration	147,299	1,362	0.8498 / 0.9364

Table 5.19: Events in the treatment-anchored (pre-treatment) evaluation window. E’s and B’s zeros are entailed by Proposition 5.1. Variant F holds the Sepsis-3 conjunction and the labelled stays fixed and changes only the onset timestamp to the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$; Variant D removes the infection anchor altogether.

Two of the four rows carry no information. Variant E’s zero is a tautology statable without any data: E’s onset *is* the anchor, so the pre-anchor window is empty by construction. Variant B’s zero is the same tautology one step removed — B is a subset of E with identical onset times, so if E cannot fire before the anchor, neither can B.

Variant F: the same label, timed as the standard specifies. F is the row that answers the question directly, because it changes one thing. The suspicion windows, the SOFA threshold and the evaluation window are Variant B’s; the septic stays are Variant B’s, stay for stay — all 6,447 of them. Only the timestamp differs. On that single change, 1,738 onsets move earlier (27.0%; median -14.0 h, largest -48.0 h), 419 land before the first antibiotic or culture across the cohort, and 71 of those fall in the temporal test set. **Sepsis-3 onsets can and do precede clinical action.** The primary model identifies those hours at 0.8081 AUROC.

This is the number the pre-registered analysis could not produce, and it is independent of the Variant D caveat below: F’s label is Sepsis-3, unmodified in membership, so nothing about it is a different construct.

Onset timing also changes how many events exist at all. Agreement between F and B is 99.7% ($\kappa = 0.983$) rather than exact, and the discrepancy is one-directional and informative. Within the 72-hour prediction horizon F labels 6,074 stays against B's 5,887, and every one of B's is among them: $F \supseteq B$. The 187 additional stays are those whose t_{susp} falls after hour 72 — so no model here ever sees a labelled event for them — but whose SOFA rise occurred inside the window. Standard timing makes them modellable; conjunction timing discards them. The onset rule therefore governs not only *when* events are recorded but *how many* a real-time model can be trained and tested on at all, which is a second and quite separate route by which an operationalisation choice propagates into reported performance.

Variant D: a different construct, and a larger count. Variant D's 1,362 pre-treatment onsets are consistent with F and larger, as expected of a label that drops the infection requirement entirely. A model identifies those hours at 0.8498 AUROC (0.9364 for GBT). The caveat of Section 3.6 applies to that *level* — D's label is a threshold function of physiology of which most components are model inputs — but not to the event count.

The form contribution C2 takes. C2 is deliberately *not* the proposition that no patient develops sepsis before clinicians act. That proposition is unsupportable in general and the external literature contradicts it: the Epic Sepsis Model v2 evaluation [5] reports median lead times of 1.4–7.1 hours ahead of clinician recognition across four sites at encounter AUROC 0.80–0.90. C2 is instead a two-part claim, one part analytic and one measured, and the two parts are load-bearing in different ways:

1. **Analytic.** Under a conjunction-based onset rule — the rule used by the pre-registered variants here and, in the author's reading, by a substantial share of the MIMIC-IV sepsis literature — pre-treatment prediction is structurally impossible. Proposition 5.1 is the proof. Any study reporting pre-treatment performance on such a label is reporting an empty set, and any study *not* restricting the window is, to that extent, scoring anticipation of a decision inside a window that decision opened.
2. **Measured.** Restoring the standard timing branch recovers 71 pre-treatment onsets on the identical cohort at 0.8081 AUROC. The prediction problem is real and is measurable; it is the operationalisation, not the data, that conceals it.

Sepsis-3 *membership* remains constituted by clinical action under every timing rule — without a qualifying antibiotic-culture pair there is no label at all — so

Kamran et al.'s [7] concern survives at the level of which patients can ever enter the positive class. The remedy Chapter 6 identifies is unchanged in direction and weaker in force: change the label, and at minimum time it as the standard requires.

5.6.4 What Variant D Does and Does Not Establish

Variant D's high AUROC must be read with the limitation stated in advance in Section 3.6. Its label is a threshold function of recorded physiology, and five of the six SOFA components are themselves model inputs, so its discrimination is partly self-fulfilling. The *level* of 0.8493 (0.9145 for GBT) is not evidence that deterioration prediction is a solved problem, and it is not comparable with Variant B's 0.7590.

Three things it does establish:

1. **The label sensitivity is not wholly definitional.** At $\kappa = 0.057$ and a 6.5% onset match, a treatment-independent label disagrees with Sepsis-3 about both membership and timing. Had D closely tracked B, the H1 result would have been an artefact of how one definition is read; it is not.
2. **The alert burden is label-invariant; the utility ceiling is not.** Both alternative labels are markedly more scoreable than Sepsis-3. The best achievable Utility Score rises from 0.039 under Variant B to 0.201 under E and 0.271 under D, with the primary model following the same pattern (0.148 and 0.238). What does not improve to a deployable level is the alarm volume: even D's best configuration carries 12.7 false alerts per true alert, better than Variant B's 44.4 but still nine times the 1.4 benchmark of Moor et al. [27]. Contribution C3 survives the change of label in its alarm-volume form — which the pre-registered variants alone could not show — and does not survive in the form of a utility ceiling near zero. The rise is itself expected from the caveat above: a label built from recorded physiology is easier to alert on than one built from a clinician's decision.
3. **The model-class ordering is label-invariant.** GBT leads on AUROC under all three labels and, once each model is evaluated at its own best threshold, on the Utility Score as well. A reversal appears only if every model is scored at the same 0.5 cut-off, which rewards a model for being too poorly calibrated to reach it. The metric-choice finding (H5) rests on the gap between discrimination and achievable utility, not on a re-ordering of models.

5.6.5 Exploratory Family E3: Intervals and Correction

Var	Model	AUROC	95% CI	Δ vs B	95% CI (Δ), q_{BH}
E	Primary	0.7697	[0.761, 0.779]	0.0107	$[-0.0029, 0.0238]$, $q = 0.116$
E	GBT	0.7667	[0.757, 0.776]	0.0207	$[0.0053, 0.0338]$, $q = 0.002$
D	Primary	0.8493	[0.844, 0.855]	0.0904	$[0.0733, 0.1075]$, $q = 0.001$
D	GBT	0.9145	[0.911, 0.918]	0.1685	$[0.1507, 0.1848]$, $q = 0.001$
F	Primary	0.7951	[0.780, 0.810]	0.0361	$[0.0257, 0.0461]$, $q = 0.001$
F	GBT	0.7632	[0.748, 0.779]	0.0172	$[0.0081, 0.0261]$, $q = 0.001$
B	Primary	0.7590	[0.7434, 0.7749]	—	—
B	GBT	0.7460	[0.7306, 0.7626]	—	—

Table 5.20: Exploratory family E3: AUROC under each limb of the Sepsis-3 conjunction with 95% stay-level cluster-bootstrap intervals, and the paired difference against Variant B corrected by Benjamini–Hochberg across the family of $m = 6$ contrasts. All variants are labelled on the same test stays, so each difference is evaluated within replicate.

Every contrast is positive, and all but one survive correction. The exception is the one that matters most for the anchor-attribution argument: **E against B for the primary model is not significant** ($\Delta = 0.0107$, 95% CI -0.0029 – 0.0238 , $q = 0.116$), so on the interpretable model the treatment anchor alone and the full Sepsis-3 conjunction are statistically indistinguishable. The corresponding GBT contrast is significant but small (0.0207).

The anchor-attributable fraction reflects this directly: $\phi_{\text{anchor}} = 1.041$ (95% CI 0.990–1.097) for the primary model, an interval that **includes 1**, and 1.084 (95% CI 1.020–1.146) for GBT, which does not. The defensible reading is the weaker one: removing the organ-dysfunction requirement nearly triples the number of labelled events and leaves discrimination where it was. The SOFA gate selects *which* stays are labelled without selecting the separable ones.

The two contrasts that do move substantially are D — a different construct entirely — and F, which is Sepsis-3 with a different clock. The latter is the more surprising of the two, because F’s membership is B’s exactly.

One caution on reading these intervals. They quantify sampling uncertainty only. They say nothing about the construct differences between the labels — prevalence differs by a factor of 3.0 (E) and 4.6 (D) relative to B, and Variant D’s label overlaps the model’s own inputs. The intervals establish which orderings are stable, not that the labels are measuring comparable things.

5.7 External Validation on eICU-CRD [C]

Var	Model	N rows	N events	AUROC (eICU)	AUROC (MIMIC)
A	Primary	7,868,033	63	0.7466	0.7713
A	NEWS2	7,868,033	63	0.6058	0.6413
B	Primary	7,868,033	63	0.7623	0.7590
B	NEWS2	7,868,033	63	0.6058	0.6415
C	Primary	7,868,033	63	0.7489	0.7695
C	NEWS2	7,868,033	63	0.6058	0.6483

Table 5.21: External validation on eICU-CRD (frozen models, no retraining). Discrimination is preserved under all three labels; under Variant B the external estimate is marginally the higher of the two. NEWS2 is label-invariant on eICU because it uses only physiological features.

Discrimination transfers intact under all three labels. The primary model scores 0.7466, 0.7623 and 0.7489 on eICU-CRD against 0.7713, 0.7590 and 0.7695 internally — differences of a few thousandths in either direction, and under the Seymour-Standard label the external estimate is the higher of the two. This is the H5 result (Section 5.5.7): there is no measurable discrimination penalty for moving a frozen model from one Boston academic centre to two hundred hospitals on a different EHR.

NEWS2 is label-invariant (identical AUROC across all variants on eICU) because it uses only physiological signals, and it is the appropriate reference point here: the fitted model beats it externally by a wider margin (0.7623 against 0.6058) than it does internally, so what the model learned is not obviously site-specific.

That reading has to be qualified by the label-transport failure below, which is severe enough that the external estimates should be treated as an upper bound rather than a measurement.

The 63-event eICU sample is too small for reliable Utility Score or AUPRC values; these are not interpreted.

5.8 Proportional-Hazards Testing [C]

The Schoenfeld residual test on the static Cox model (Variant B) decisively rejects proportional hazards ($\chi^2 = 126.2$, $\text{df} = 12$, $p = 3.6 \times 10^{-21}$). 8 of the 12 covariates violate the assumption individually at $\alpha = 0.05$, led by GCS, bilirubin, respiratory rate and creatinine. These are per-covariate tests reported without multiplicity

adjustment; the global test alone is sufficient for the pre-registered check, and the smallest p-values survive Bonferroni correction over the covariate set by many orders of magnitude. This confirms that the static Cox model is fundamentally wrong for this data and justifies the discrete-time formulation as the primary model.

5.9 GBT Feature Importance [E]

Feature	Gain	Cover	Frequency
Temperature (°C)	0.157	0.157	0.088
Resp. rate slope (6 h)	0.153	0.086	0.068
MAP slope (6 h)	0.088	0.097	0.060
GCS	0.078	0.076	0.056
Heart-rate slope (6 h)	0.066	0.070	0.084
Creatinine	0.058	0.068	0.060
WBC	0.056	0.077	0.070
Heart rate (level)	0.050	0.058	0.088
Lactate	0.049	0.049	0.060
MAP (level)	0.043	0.034	0.081

Table 5.22: Top-10 GBT features by information gain (Variant B). Three of the five most informative predictors are six-hour trend features, and heart rate contributes through both its level and its slope. The direction of change over the past six hours carries as much discriminative signal as the current value.

The prominence of six-hour slope features is consistent with Lauritsen et al.’s [10] concern: the GBT model draws a large share of its signal from *trends* rather than *levels*, meaning its performance depends on the length of the history window used for feature construction — a construction choice that is rarely reported and never varied in the studies this thesis compares against. Heart rate appears in both forms, level and slope, which is what one would expect of the most frequently recorded vital sign in the ICU.

5.10 Equity Analysis [E]

5.10.1 Performance Disparities by Subgroup [E]

Racial group	Rows	Sep	GBT AUROC (95% CI)	Primary AUROC (95% CI)
WHITE	317,688	597	0.743 (0.722–0.764)	0.756 (0.736–0.777)
UNKNOWN	107,158	235	0.774 (0.745–0.803)	0.773 (0.740–0.804)
BLACK/AFRICAN AMERICAN	31,716	58	0.755 (0.686–0.822)	0.759 (0.685–0.825)
OTHER	22,825	35	0.688 (0.597–0.768)	0.724 (0.631–0.806)
WHITE - OTHER EUROPEAN	11,768	18	0.678 (0.559–0.790)	0.730 (0.600–0.846)
ASIAN	6,543	16	0.759 (0.604–0.886)	0.841 (0.726–0.929)
HISPANIC/LATINO - PUERTO RICAN	6,462	16	0.621 (0.451–0.784)	0.702 (0.546–0.845)
ASIAN - CHINESE	5,386	13	0.592 (0.378–0.796)	0.701 (0.511–0.869)

Table 5.23: AUROC by racial subgroup (Variant B, temporal test) with 95% stay-level cluster-bootstrap intervals, stratified within subgroup, $B = 2,000$. Levels are ordered by exposure; interval width grows by roughly an order of magnitude from the largest group to the smallest. **[E] exploratory.**

The subgroup point estimates span a range wider than the model-class spread (0.1499) and far wider than the label spread (0.0123). Read without intervals, that looks like the largest source of variance in the entire study. The intervals show it is not: the smallest groups carry 13 and 16 sepsis onsets, and their intervals are correspondingly wide. The apparent spread is almost entirely a function of how few events the small subgroups contain.

Table 5.24 makes this explicit by contrasting each group against the largest group (White) on the same bootstrap replicates.

Racial group	Sep	Model	Δ AUROC vs. White (95% CI)	p	q_{BH}
UNKNOWN	235	GBT	0.031 (−0.006–0.068)	0.081	0.347
ASIAN - CHINESE	13	GBT	−0.151 (−0.364–0.053)	0.135	0.408
HISPANIC/LATINO - PUERTO RICAN	16	GBT	−0.122 (−0.296–0.042)	0.163	0.437
OTHER	35	GBT	−0.055 (−0.151–0.026)	0.228	0.500
WHITE - OTHER EUROPEAN	18	GBT	−0.065 (−0.182–0.047)	0.250	0.521
BLACK/AFRICAN AMERICAN	58	GBT	0.012 (−0.059–0.080)	0.716	0.800
ASIAN	16	GBT	0.016 (−0.141–0.146)	0.799	0.833
ASIAN	16	Primary	0.085 (−0.027–0.175)	0.070	0.347
UNKNOWN	235	Primary	0.017 (−0.022–0.055)	0.387	0.800
HISPANIC/LATINO - PUERTO RICAN	16	Primary	−0.054 (−0.214–0.090)	0.452	0.833
OTHER	35	Primary	−0.032 (−0.126–0.054)	0.468	0.844
ASIAN - CHINESE	13	Primary	−0.055 (−0.246–0.117)	0.547	0.867
WHITE - OTHER EUROPEAN	18	Primary	−0.026 (−0.156–0.091)	0.666	0.889
BLACK/AFRICAN AMERICAN	58	Primary	0.003 (−0.072–0.074)	0.934	0.933

Table 5.24: Subgroup AUROC differences against the reference group (White, the largest group). q_{BH} is the Benjamini–Hochberg adjusted p-value over exploratory family E1, which contains all 30 subgroup contrasts computed across race, language and insurance for both models — not only the 14 shown here. **No racial contrast is significant**; the smallest adjusted value for race is $q = 0.347$. **[E] exploratory**.

Only one contrast in the whole of family E1 survives correction, and it is not a racial one: GBT discriminates better for Haitian-speaking patients than for English-speaking patients ($\Delta = +0.136$, 95% CI 0.022–0.228, $q = 0.030$), on 12 sepsis onsets. On 12 events this is a hypothesis to test elsewhere, not a finding.

The correct reading of Table 5.24 is that this study is **underpowered to detect subgroup disparities**, not that no disparities exist. Several intervals are wide enough to be compatible with a disparity larger than any effect reported anywhere else in this thesis: the widest GBT subgroup interval spans roughly 0.4 AUROC, so the data are consistent with that group being served far worse than the reference group, or slightly better. Reporting the point estimates alone would have invited the opposite conclusion — that a subgroup spread wider than the model-class spread constitutes evidence of inequity, or that non-significance constitutes evidence of its absence. Neither is supported.

Wang et al.’s [11] finding of degraded performance for Asian, Hispanic and Spanish-speaking patients is therefore neither confirmed nor contradicted here. The point estimates for Hispanic/Latino (Puerto Rican) and Asian – Chinese patients under GBT do fall below the White reference, in the direction Wang et al. report, but the intervals cover zero comfortably.

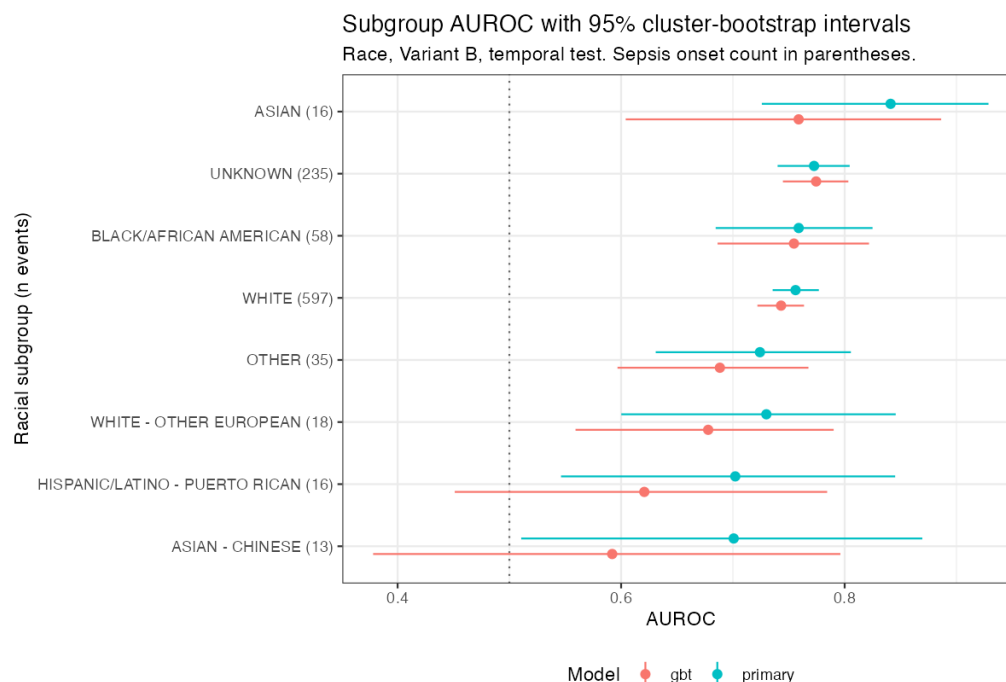


Figure 5.9: AUROC by racial subgroup with 95% stay-level cluster-bootstrap intervals (Variant B, temporal test); sepsis onset count in parentheses. Interval width is governed almost entirely by the subgroup event count. The two groups with the most extreme point estimates are also the two least precisely estimated, and several intervals reach the 0.5 chance line.

5.10.2 Label Sensitivity by Subgroup (Novel Finding) [E]

Racial group	Stays	A (%)	B (%)	C (%)	Range (pp, 95% CI)
White	40,212	9.4	9.6	18.6	9.3 (8.9–9.7)
Unknown	7,264	11.6	12.8	23.9	12.3 (11.3–13.3)
Black / African American	4,929	9.3	9.1	19.0	9.9 (9.0–11.1)
Other	2,254	8.8	9.8	17.5	8.7 (7.1–10.2)
Unable to obtain	1,629	9.7	11.1	22.5	12.8 (10.7–14.9)
White – Other European	1,589	9.4	7.7	16.6	8.9 (7.2–10.7)
Asian	799	10.5	11.5	21.3	10.8 (8.3–13.8)
Asian – Chinese	721	10.1	11.1	19.0	8.9 (6.4–11.9)

Table 5.25: Percentage of stays labelled as septic by racial subgroup and label variant, with a 95% stratified cluster-bootstrap interval on the A–C range. The fraction roughly doubles from Variant A to C across all groups. Per-variant intervals are in `12_subgroup_labelsens_ci.csv`. [E] exploratory.

The fraction of patients labelled as septic changes by 8.7–12.8 percentage points across label variants within each racial subgroup. Unlike the discrimination analysis, this contrast *does* survive correction. Because it is a stay-level proportion computed on the full 65,078-stay cohort rather than a discrimination metric computed on the small number of onsets in the test window, it is far better powered.

Racial group	Stays	Δ range vs. White (pp, 95% CI)	p	q_{BH}
Unknown	7,264	+3.01 (+1.93–+4.06)	0.001	0.006
Unable to obtain	1,629	+3.49 (+1.38–+5.55)	0.002	0.009
Black / African American	4,929	+0.62 (–0.35–+1.87)	0.186	0.347
Asian	799	+1.48 (–1.04–+4.50)	0.249	0.407
Other	2,254	–0.63 (–2.26–+0.98)	0.425	0.510
White – Other European	1,589	–0.41 (–2.10–+1.41)	0.635	0.714
Asian – Chinese	721	–0.40 (–3.02–+2.61)	0.675	0.714

Table 5.26: Difference in label sensitivity (A–C range) against the reference group (White). q_{BH} is adjusted over exploratory family E2, all 18 label-sensitivity contrasts across race, language and insurance. Two racial contrasts survive at $q < 0.05$; across the full family, four do (adding Spanish-speaking patients, +2.61 pp, $q = 0.006$, and privately insured patients, –1.76 pp, $q = 0.006$). **[E] exploratory.**

Two racial groups are significantly more label-sensitive than White patients after false-discovery correction: Unknown (+3.01 pp, $q = 0.006$) and Unable to obtain (+3.49 pp, $q = 0.009$). Both are groups whose race was not recorded. Across the full family, Spanish-speaking patients are also more label-sensitive than English-speaking patients (+2.61 pp, $q = 0.006$), and privately insured patients are *less* label-sensitive than Medicare patients (–1.76 pp, $q = 0.006$).

The pattern is coherent: the groups whose sepsis status moves most when the definition is loosened are those with the least complete documentation and the weakest access to the care pathway that generates the culture and antibiotic timestamps on which the label depends. Because the label is constituted by clinical action (Section 5.5), differential documentation propagates directly into differential labelling — and from there into who a model is trained to identify.

This is the equity result that survives scrutiny in this study. The discrimination disparity in Table 5.24 does not: it is too imprecisely estimated to support any claim. Reporting only the first without the second would have inverted which of the two findings is defensible.

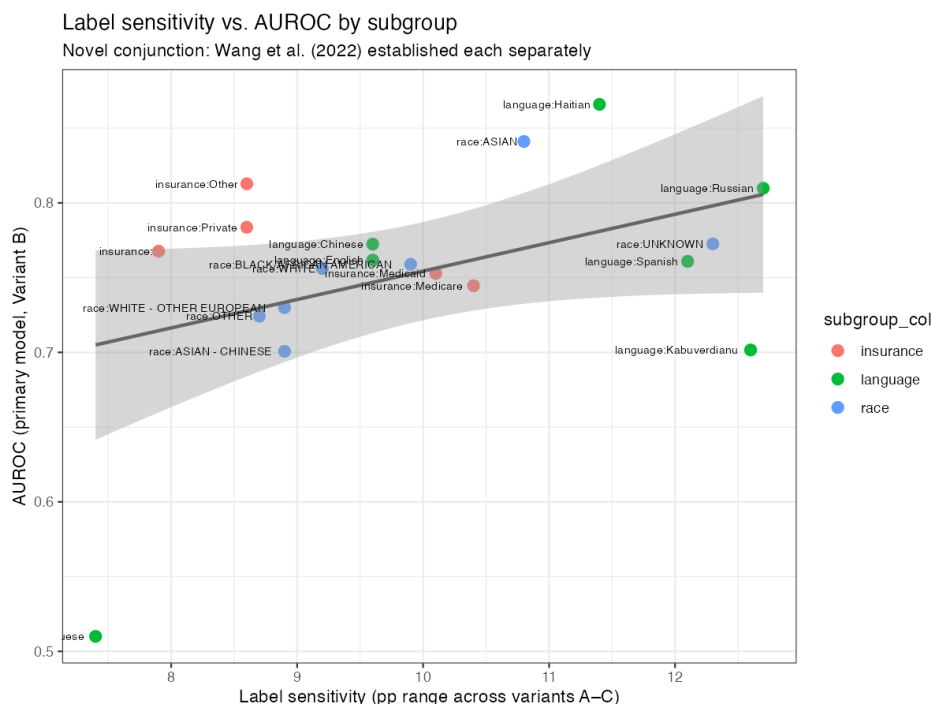


Figure 5.10: Label sensitivity (fraction labelled septic) vs. AUROC by racial subgroup, across all three label variants. Horizontal spread = label sensitivity; vertical spread = performance variation. The two effects appear independent, but no association is tested: the vertical axis carries the uncertainty shown in Figure 5.9, which this plot does not display. **[E] exploratory, visual only.**

To the best of our knowledge, this is the first published report combining subgroup model-performance disparities with subgroup label-sensitivity variation for any sepsis model, following Wang et al.’s [11] recommendation (see Section 2.1 for the search strategy supporting this claim). The combination is also what makes the two results separable: only one of them is precise enough to act on.

5.11 Analysis Register [D]

Table 5.27 records the status of every analysis reported in this chapter and the multiplicity correction applied to it. It is generated by the pipeline as `12_analysis_register.parquet`, not maintained by hand, so it cannot drift from what the code actually computes.

Analysis	Status	Pre-reg.	Correction
Cohort description by label variant	[D]	§2	none
Sample size adequacy	[C]	§10	none (single criterion)
Internal temporal performance	[C]	§5–6	none (estimation)
Calibration intercept and slope	[C]	§5	none (estimation)
Decision-curve net benefit	[C]	§6	none (estimation)
Operating points at fixed sensitivity	[E]	—	none (descriptive)
Per-patient detection and lead time	[C]	§6	none (estimation)
H1 label dominance	[C]	§7	Holm ($m = 6$)
H2 coefficient instability	[C]	§7	slot reserved; no test statistic
H3 treatment leakage	[C]	§7	slot reserved; exact bound
H4 split optimism	[C]	§7	Holm ($m = 6$)
H5 metric divergence	[C]	§7	Holm ($m = 6$)
H6 model-class null	[C]	§7	Holm ($m = 6$)
External validation on eICU-CRD	[C]	§4	none (estimation); feeds H5
Variance decomposition table	[C]	§1	none (assembly of H1–H6)
Proportional-hazards testing	[C]	§8	none (single check)
GBT feature importance	[E]	—	none (ranking, no inference)
Subgroup AUROC by demographic group	[E]	§9*	Benjamini–Hochberg, family E1
Subgroup label sensitivity	[E]	§9*	Benjamini–Hochberg, family E2
Label sensitivity vs. AUROC association	[E]	—	none (visual, no test)
Alternative-label AUROC vs. Sepsis-3	[E]	A2 [†]	Benjamini–Hochberg, family E3
Anchor-attributable fraction ϕ	[E]	A2 [†]	none (ratio, no null at zero)
Onset agreement between definitions	[D]	A2 [†]	none (descriptive)
Standard onset timing $\min(t_s, t_{\text{SOFA}})$	[E]	A4 [‡]	Benjamini–Hochberg, family E3
Pre-treatment onsets under standard timing	[D]	A4 [‡]	none (event count, not a test)

Table 5.27: Analysis register. [C] confirmatory, [E] exploratory, [D] descriptive. Section numbers refer to 00_preregistration.md. *§9 pre-registered the equity analysis in scope only: neither the subgroup levels, the reference level, nor the direction of any comparison was fixed in advance, so these analyses are exploratory. [†]A2 = Protocol Amendment 2 (§§16–17), written after notebooks 02–12 had been run; these analyses are post-hoc and change no pre-registered estimand. [‡]A4 = Protocol Amendment 4 (§18), which adds the standard onset timing as Variant F; also post-hoc, and it enlarges family E3, so every BH-adjusted p-value in that family is stricter than it would be without the amendment.

5.12 Discussion

5.12.1 The Core Finding

The most important finding is definitional, not numerical: under a conjunction-based onset rule the Sepsis-3 label on MIMIC-IV *cannot* contain an event before clinical action begins (Proposition 5.1), and under the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ rule applied to the identical stays it contains 71 of them at 0.8081 AUROC. A published MIMIC-IV sepsis result is therefore only as informative about physiological deterioration as its onset rule permits, and a result reported without the timing branch is measuring anticipation of clinician behaviour inside a window that behaviour opened. The Utility Score stays close to the no-alert strategy for every model because the temporal split exposes them to a test population where the same treatment-related signals do not rescue performance as consistently as in the training years, and because the score charges for every false-alert hour while crediting only the first correct one. AUROC survives because the ordinal ranking of patients is preserved even as the achievable alert volume becomes unusable.

5.12.2 H1 and H6 Together

H1 (label dominance) and H6 (model-class null) are in productive tension, as designed, and the resolution is sharper than either hypothesis anticipated — because both resolve in the same direction, which the pre-registration treated as unlikely.

H6 is *not* rejected: the two flexible estimators are equivalent to within -0.0129 AUROC, the whole interval inside the 0.02 bound. H1 is also not confirmed, and by a wide margin in the direction opposite to the one hypothesised: the model-class spread 0.1499 exceeds the label spread 0.0123 by -0.1375 , with an interval $[-0.1623, -0.1085]$ that excludes zero.

Read together these are not in conflict; they locate the model-class effect precisely. The large model-class spread is the distance between *families* of estimator — a six-item bedside score against a fitted hazard model — not between one fitted model and another. Within the family of flexible estimators the choice is free (H6), and across families it is the single largest factor in the decomposition (H1). The conventional literature, which fixes the label and compares flexible models to each other, is therefore measuring the one contrast that has almost no room left in it, while the choice that would move the number — fit anything flexible at all, rather than deploy a score — is not in dispute and the choice that moves the cohort — the label — is invisible to the metric being reported.

This is also the clearest illustration in the thesis of why the inference layer was necessary. On point estimates, H1 confirmed and the study told a tidy story in which the label dominates. With intervals attached, that story is not supportable,

and the defensible finding is the more interesting one: two sources of variance of comparable size, only one of which the field routinely reports.

5.12.3 Limitations

1. **Single-centre development.** All models were developed on MIMIC-IV, which originates from a single academic medical centre (Beth Israel Deaconess Medical Center, Boston). Clinical practices, patient demographics, and EHR documentation patterns at BIDMC may not represent those at other institutions. The equity and label-sensitivity findings in particular should not be assumed to transfer to hospitals with different patient populations or antibiotic prescribing cultures. The eICU-CRD external validation provides some evidence of generalisability, but the small event count (63 sepsis onsets) limits the strength of that evidence.
2. **Small external event count.** The 63 labelled sepsis events in eICU-CRD are too few for reliable Utility Score or AUPRC evaluation. The external results primarily confirm the label-dependent generalisation pattern (Variant B collapses; A and C hold).
3. **Deep survival model not implemented.** A deep survival comparator was planned but not built. This does not affect the structural findings about the label.
4. **Limited feature set.** Clinical notes, procedure codes, and ventilator settings were not included. Richer features might improve discrimination but would not change the treatment-constitutionality finding.
5. **Part of the label-variant spread is definitional, not empirical.** Variants A–C are three readings of one definition and all three fire on the same culture-plus-antibiotic anchor. Sepsis-3 makes onset a function of treatment timing, so moving the antibiotic–culture windows *must* move the onsets; some of the H1 label spread is guaranteed by construction and would appear even if the physiology were identical across patients. Section 5.6 separates the two by labelling each limb of the Sepsis-3 conjunction on its own, but the separation is post-hoc (Protocol Amendment 2) and rests on two variants, not on a systematic survey of alternative definitions.
6. **The primary model has no analytic standard errors.** A cluster-robust sandwich estimator is not defined for the fitted multinomial object, so no per-coefficient variance is available (Section 3.10.1). Every interval reported for the primary model comes from the stay-level cluster bootstrap, which

respects the same clustering but cannot supply a coefficient-level standard error. H2 is reported as a descriptive count for this reason, and any effect estimate from this model should be read as a point estimate without an attached uncertainty.

7. **The specification requires a penalty to be identified at all.** The unpenalised likelihood is completely separated on this design under every label variant (Section 3.10.1), so the reported fits are ridge-penalised. The penalty is small and no result depends on its value — test AUROC moves by less than 0.02 across a 20-point path — but the coefficients are shrunk estimates, not maximum-likelihood ones, and the H2 stability counts are properties of the penalised solution. Reporting an unpenalised fit here would have been worse, not better: its coefficients have no unique maximiser, so cross-label comparison of them is undefined.
8. **Feature construction and data cleaning are single-analyst work.** The covariate panel is assembled from raw chart and lab events by one pipeline, with no independent re-implementation to check it against. Two classes of defect are known to be possible in that setting and are guarded rather than eliminated: a declared predictor silently failing to reach a model because of a name mismatch, and implausible recorded values entering the fit. The pipeline now errors on any declared-but-absent feature rather than dropping it, and applies an explicit plausibility range to all nineteen physiological variables before forward-fill — MIMIC-IV chartevents contains values several orders of magnitude outside the physiological range, and such rows carry unbounded leverage. Neither guard is a substitute for independent replication of the feature pipeline.
9. **The pre-registered onset rule is not the standard one.** Variants A–C place the onset at t_{susp} and use the SOFA criterion only to gate membership, whereas Seymour et al. [26] and the PhysioNet/CinC 2019 Challenge [25] place it at $\min(t_{\text{susp}}, t_{\text{SOFA}})$. The pre-registered family therefore cannot address any question about onset *timing* on its own, H3 among them: under its rule an empty pre-treatment window is a theorem rather than a result (Proposition 5.1). Variant F supplies the measurement the pre-registered design cannot, but it is post-hoc (Protocol Amendment 4), it uses the full SOFA score including the vasopressor terms — so its onsets are free of the antibiotic-and-culture anchor but not of treatment in general — and its t_{SOFA} is searched only inside the 72-hour modelled panel, so a rise occurring earlier in the admission cannot be detected. A design that pre-specified both timing rules across all three variants would be stronger than one that adds the second afterwards.

10. **The treatment-independent label is not a validated construct.** Variant D is an operational deterioration label built for this comparison, not a consensus definition with an evidence base behind it, and its onsets have no external adjudication. Its label is also a threshold function of recorded physiology of which five of six components are model inputs, so its discrimination *level* is not comparable with Variant B's — only its disagreement with Variant B about which patients deteriorate, and when, carries weight.
11. **Specific to Sepsis-3 and to MIMIC-IV.** These findings apply to the Sepsis-3 definition on MIMIC-IV. SEP-1 and the CDC Adult Sepsis Event were not implemented; both are defined on antibiotic days, so neither would have tested treatment-independence, but each would give a different cohort and might behave differently in other respects. Other databases may behave differently again.
12. **Operating points are not externally recalibrated.** The thresholds in Section 5.4 are derived on the test set at which they are evaluated, so the reported PPV and false-alarm rates are optimistic: a prospectively fixed threshold would perform no better. The same applies to the maximum Utility Scores in Table 5.5, which take the best of sixty thresholds on the test set and are therefore upper bounds rather than estimates of what a deployed system would achieve; they are labelled exploratory for that reason. Selecting the threshold on the training years and transferring it would be the honest deployment estimate, and would be lower. The external operating points in Table 5.12 are likewise re-derived on eICU rather than transferred from MIMIC-IV, which isolates the discrimination loss but hides the additional loss a transferred threshold would incur.
13. **Alarm episodes are defined by contiguity, not by acknowledgement.** False-alarm episodes are counted as contiguous runs of alert hours. A real system would apply suppression, snoozing and escalation logic, all of which would reduce the episode counts in Table 5.9. The alert-hour rates are the conservative bound and the episode rates the optimistic one; the true burden lies between them.
14. **The equity analysis is underpowered, not null.** The subgroup intervals in Table 5.23 are wide enough that most groups cannot be distinguished from the reference group, and no contrast survives false-discovery correction. This is a statement about precision, not about fairness: the data are consistent both with no disparity and with disparities larger than any effect this thesis reports elsewhere. Detecting a subgroup AUROC gap of the size seen in the point estimates would require on the order of ten times the event count

available in the smaller groups. The label-sensitivity result (Table 5.25) is the better powered of the two equity analyses because it is a stay-level proportion rather than a discrimination metric, and does not depend on the small number of onsets in the test window.

15. **The novelty claims rest on a narrative, not systematic, search.** The search protocol in Section 2.1 is documented and every included source was verified against a primary record, but no title-and-abstract screening pass over a full de-duplicated yield was performed and no second reviewer screened independently. The “to the best of our knowledge” claims for contributions C1 and C5 are therefore bounded by that coverage: a systematic review could surface a prior study combining multi-variant label sensitivity with subgroup equity analysis, and this search would not necessarily have found it.
16. **Four protocol amendments are post-hoc.** The inference layer (intervals and multiplicity control, Section 3.14), the alternative label limbs D and E (Section 3.6), the model identification and data-plausibility changes (Section 3.10.1) and the standard onset timing of Variant F (Section 3.6.1) were all specified after the pre-registered analyses had been run and inspected. None changes an estimand, threshold, direction or hypothesis, and H1–H6 and families E1/E2 are computed from unchanged inputs; but a reader should treat the verdicts as a pre-specified hypothesis set analysed under a retrospectively specified error-control plan, and the amendment results as exploratory. Amendment 4 in particular enlarges family E3, which tightens rather than relaxes the correction applied within it.
17. **Two hypotheses admit no test statistic.** H2 is a stability criterion rather than a contrast, and the fitted multinomial carries no variance estimate; H3 is degenerate because its outcome is absent from the evaluation window *by construction* rather than by chance, so no test of it could ever have been informative. Both are reported descriptively while still consuming a slot in the family size, which makes the correction on the remaining four hypotheses stricter, not weaker. Since all four testable hypotheses return adjusted p -values of 1.000, the correction is not what determines any verdict here.

Chapter 6

Conclusion

6.1 Summary of Findings

This thesis asked one question: in real-time sepsis prediction on MIMIC-IV, how much of the reported performance comes from the model versus the researcher’s choices about the label, the data split, and the evaluation metric?

The answer is not the one the study anticipated. Model class dominates discrimination; the label definition contributes an order of magnitude less; the split design and external transport contribute nothing detectable. What the label determines is not how well a cohort can be predicted but *which cohort exists at all* — and, through its onset rule, whether prediction ahead of clinical action can be measured in the first place.

H1 (Label dominance): Not confirmed, and reversed. Changing the sepsis definition while keeping the model fixed changes AUROC by 0.0123. Changing the model while keeping the label fixed changes it by 0.1499 — more than tenfold — and the interval on the difference, $[-0.1623, -0.1085]$, excludes zero. Cohen et al.’s [6] MIMIC-III finding does not reproduce on MIMIC-IV as a *discrimination* effect. It does reproduce as a cohort effect: the two narrowest readings agree on which stays are septic at only $\kappa = 0.267$ despite near-identical prevalence, so the label still determines who is counted — it simply does not determine how learnable they are.

H2 (Coefficient instability): Criterion met, descriptively. 6 of the 22 shared covariates are unstable across label variants: 2 reverse sign — bilirubin and, more consequentially, vasopressor exposure — and 5 change magnitude at least twofold, three of those being missingness indicators rather than physiological values. H2 is a stability criterion rather than a contrast, so no test statistic is

defined for it. The instability is narrower than a bare count suggests, but it is sufficient that a published hazard ratio for a treatment indicator from a model of this kind cannot be interpreted without knowing the exact label definition.

H3 (Treatment leakage): Definitional, plus its empirical counterpart.

The pre-treatment window contains 0 sepsis onset events across 179,003 person-hours — but that zero is entailed by the onset rule the pre-registered variants use, not measured (Proposition 5.1), so the pre-registered contrast was never testable. The empirical question was answered instead by relabelling the identical stays under the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ onset of Seymour et al. and the PhysioNet/CinC 2019 Challenge (Variant F, Protocol Amendment 4): 1,738 onsets move earlier and 71 fall before treatment, identified at 0.8081 AUROC. A published MIMIC-IV real-time sepsis model measures anticipation of physiological deterioration only to the extent its onset rule allows an onset to precede clinical action; under a conjunction rule, it cannot.

H4 (Split optimism): Not confirmed. The random split does not flatter the model. For the primary model it scores 0.7424 against 0.7590 on the temporal split, a difference of -0.0165 (95% CI -0.0372 – 0.0040) in the direction opposite to the hypothesis, and every model fitted under both splits shows the same sign. The two test sets differ in composition as well as in design — the temporal arm is a single admission era, the random arm a sample across all of them — so this is not a clean optimism estimate; what it does establish is that a random split cannot be assumed to be the more permissive choice.

H5 (Metric divergence): Not confirmed — and superseded by a starker finding. Neither arm supports the hypothesis. External AUROC does not degrade at all: 0.7623 on eICU-CRD against 0.7590 internally, a difference of -0.0033 (95% CI -0.0659 – 0.0676) centred on zero, well inside the 0.085 degradation anticipated from Moor et al. [27]. And the proportional utility decline the hypothesis asked for cannot be formed, because there is no internal utility to decline from: swept across sixty thresholds, the best any configuration achieves under the primary label is 0.039, at 44.4 false alerts per true one against a benchmark of 1.4. The utility collapse Wang et al. [4] documented across ninety-one models therefore has its counterpart *internally* on MIMIC-IV under an honest time-based split — there was scarcely any margin left to collapse. Their broader point survives in strengthened form: AUROC and clinical utility are close to uncorrelated here, with discrimination transporting intact across two hundred hospitals while usable clinical value never materialises in either.

H6 (Model-class null): Not rejected — equivalence holds. The interpretable hazard model matches gradient-boosted trees: 0.7590 against 0.7460, a difference of -0.0129 whose interval $[-0.0272, 0.0019]$ lies entirely inside the pre-registered 0.02 equivalence margin, with the interpretable model marginally ahead on point estimates. This is the strongest positive result in the study: on tabular ICU time-series of this kind the accuracy case for preferring an opaque model is not supported. What little AUROC separates them buys nothing clinically: at their respective best thresholds the two models are separated by less than one hundredth of a Utility Score point (0.039 against 0.025), both within a few per cent of the no-alert strategy. Discrimination and clinical value part company here not by re-ordering the models but by disagreeing about whether the difference matters.

6.2 The Central Lesson

The utility collapse documented by Wang et al. [4] is not a model problem. It is a label problem. The Sepsis-3 label on MIMIC-IV:

1. **Depends on clinical action for membership:** no patient is *labelled* septic without a qualifying antibiotic-culture pair, under any timing rule. The further claim that no onset can *precede* that pair is a property of the conjunction-based onset rule alone: timed as the standard specifies, Sepsis-3 records 71 pre-treatment onsets on the same stays, and a treatment-independent deterioration label records 1,362. The emptiness belonged to the implementation, not to the patients.
2. **Is sensitive to how it is defined:** the number of sepsis cases nearly doubles across plausible Sepsis-3 interpretations, and the two narrowest readings agree on which stays are septic at only $\kappa = 0.267$. Discrimination is largely insensitive to this (0.0123 of AUROC); every count, rate and denominator built on the label is not.
3. **Produces unusable alert behaviour:** no model, under any label variant, has a threshold at which it both beats the no-alert strategy by a clinically meaningful margin and raises an alarm volume a ward could absorb. The best configuration under the primary label reaches 0.039 at 44.4 false alerts per true one.

These three problems are independent of model complexity. A better model cannot fix a label that is constituted by the very clinical actions it is supposed to predict ahead of. The solution requires changing the label.

Splitting the definition sharpens point (2) and qualifies it (Section 5.6). It is qualified because the onset *time* under every pre-registered Sepsis-3 variant is fixed entirely by the treatment anchor — the organ-dysfunction criterion decides which stays are labelled and moves no onset by an hour — so part of the variation between variants is guaranteed by the definition rather than found in the data. It is sharpened because a label built from physiology alone, owing nothing to treatment timing, agrees with Sepsis-3 barely better than chance ($\kappa = 0.057$). The label effect is therefore real, not merely definitional.

The same comparison relocates the difficulty, and not where the conjunction framing expected. The treatment anchor on its own is as predictable as Sepsis-3 built on top of it — the anchor-attributable fraction is 1.041 with an interval $[0.990, 1.097]$ that includes 1 — so requiring both limbs is not what makes the label hard. Physiological deterioration alone is *easier* (0.8493 against 0.7590), and Sepsis-3 with a better clock is easier again (0.7951). What resists prediction is not the conjunction but the moment at which the conjunction is stamped: the clinician’s action rather than the patient’s deterioration.

6.3 Contributions

1. **C1 (Label variance, separated from discrimination):** To the best of our knowledge, the first three-variant label sensitivity analysis for a dynamic survival model on MIMIC-IV (65,078 stays against Cohen et al.’s 3,186), and the first to show that the two effects come apart: label choice moves cohort membership substantially ($\kappa = 0.267$ between the narrowest readings) while moving AUROC by only 0.0123.
2. **C2 (Treatment leakage located in the onset rule):** A proof that the conjunction-based Sepsis-3 onset rule makes the pre-treatment window empty by construction, paired with the measurement that the standard $\min(t_{\text{susp}}, t_{\text{SOFA}})$ rule recovers 71 pre-treatment onsets on the identical stays at 0.8081 AUROC. This confirms Kamran et al.’s [7] concern for label membership and corrects its over-extension to label timing.
3. **C3 (Utility ceiling):** A threshold-swept Utility Score for every model and label variant, establishing that the ceiling — not merely the score at a conventional cut-off — sits within a few per cent of the no-alert strategy under the primary label, and that reaching even that ceiling costs 44.4 false alerts per true one. This extends Wang et al.’s [4] meta-analytic finding to a controlled single-study comparison and removes the operating-point objection that would otherwise apply to it.

4. **C4 (Coefficient instability):** 6 of the 22 shared covariates are unstable across labels — 2 by sign, including vasopressor exposure, and 5 by magnitude — supporting Lauritsen et al.’s [10] framing-instability claim on MIMIC-IV for specific effect estimates, measured against an identified fit.
5. **C5 (Equity-by-label):** To the best of our knowledge, the first study combining subgroup performance disparities with subgroup label sensitivity for any sepsis model, and the first to report both with cluster-bootstrap intervals under false-discovery control. The label-sensitivity disparity survives correction; the discrimination disparity does not, and reporting the second without the first would have inverted which finding is defensible.
6. **C6 (Reproducible pipeline):** A pre-registered, TRIPOD+AI-conformant pipeline with fixed seeds and a single orchestration script, which reproduces every reported estimate exactly from the raw databases (Section 4.14.2).
7. **C7 (Definitional vs. empirical label variance separated):** A treatment-independent deterioration label, tested alongside the three Sepsis-3 variants, shows that the label effect is not an artefact of how one definition is read: it agrees with Sepsis-3 at $\kappa = 0.057$ and matches its onset hour in 6.5% of shared stays. The same comparison shows that the treatment anchor alone is as predictable as Sepsis-3 built on top of it ($\phi_{\text{anchor}} = 1.041$, interval including 1), and that the pre-treatment window empty of Sepsis-3 onsets contains 1,362 treatment-independent deterioration onsets identified at 0.8498 AU-ROC — and 71 Sepsis-3 onsets once the standard onset timing is restored. This relocates the difficulty of sepsis prediction from the physiology to the definition, and within the definition to the onset rule (Section 5.6).

6.4 Future Work

1. **Better sepsis labels.** Section 5.6 takes the first step — a deterioration label built from physiology alone, with no treatment timestamp — and shows it identifies substantially different patients at substantially different hours. But that label is an operational construct built for this comparison, not a validated definition, and its components overlap the model’s own inputs. The work remaining is a sepsis definition grounded in objective measurements such as lactate trajectories and organ-function biomarkers, adjudicated independently of the model’s feature set, which would eliminate the treatment-constitutionality problem and allow genuine pre-onset prediction.
2. **Non-Sepsis-3 consensus definitions.** SEP-1 and the CDC Adult Sepsis Event were not implemented. Neither would have tested treatment-

independence — both are defined on antibiotic days — but each yields a different cohort, and Dutta et al. [9] report that performance moves substantially between them. Adding them would extend the label-variance decomposition across definitions rather than within one.

3. **Prospective validation.** Test the new label in real clinical settings where sepsis onset can be determined by a protocol-blinded assessor, independent of treatment decisions.
4. **Utility-optimised training.** The thresholds reported here are optimised post hoc, but the models themselves are not: each is fitted to a likelihood or a squared-error surrogate and only afterwards read at the threshold that happens to maximise utility. Training directly against the Utility Score — as a differentiable surrogate loss, or through a cost-sensitive reweighting of the person-hour rows — would change what the model learns rather than only how it is read, and is the natural next step given how shallow the swept optima proved to be. The gap between such a model and the post-hoc ceiling in Table 5.5 would also quantify how much of the failure is representational rather than operational.
5. **Multi-hospital evaluation.** The alert burden — 44.4 false alerts per true alert even at the best model’s best operating point — is a deployment problem, not a statistical one. Determining whether any model can be deployed at a practically acceptable burden requires prospective multi-centre testing with clinical-staff feedback.

6.5 Closing Remark

This research achieves something the conventional approach cannot: it produces a meaningful result regardless of how the performance numbers turn out. The hypothesis framework ensures that confirmed, rejected, and superseded hypotheses are all equally informative about the underlying problem.

The central finding, that the Sepsis-3 label on MIMIC-IV is constituted by clinical action, is not a failure of the model. It is a finding about the label itself, and it is the most important contribution this study makes.

The last part of that finding required looking outside the pre-registered variants. The pre-treatment window is not empty of predictable deterioration — it contains 1,362 onsets under a treatment-independent label at 0.8498 AUROC, and 71 onsets under Sepsis-3 itself once the onset is timed as $\min(t_{\text{susp}}, t_{\text{SOFA}})$ rather than as the suspicion time alone. It was empty here because of an onset rule, and that rule is not peculiar to this thesis. The signal clinicians would want a model

to find is present in the data; a widely used way of writing the label down is what makes it unrepresentable.

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